

Billboard Hot 100 Data

Imports

```
In [147]: import pandas as pd
import matplotlib.pyplot as plt
import os
import requests
import time
import re
from difflib import SequenceMatcher
```

read in (read and then filter ≥ 2005)

```
In [8]: # read Billboard Hot 100 data
df_billboard = pd.read_csv("hot100.csv")

# convert chart dates into datetime
df_billboard["Date"] = pd.to_datetime(df_billboard["Date"])

# keep broad project time frame
df_billboard = df_billboard[
    df_billboard["Date"].dt.year >= 2005
]

latest_date = df_billboard["Date"].max()

print("Oldest date:", df_billboard["Date"].min())
print("Newest date:", latest_date)
print("Rows:", df_billboard.shape[0])
```

```
Oldest date: 2005-01-05 00:00:00
Newest date: 2026-05-20 00:00:00
Rows: 111500
```

Collapse + Build Features (chart life)

```
In [10]: df = df_billboard.copy()

# Collapse weekly chart rows into one row per Song + Artist
song_life = (
    df.sort_values("Date")
      .groupby(["Song", "Artist"])
      .agg(
          first_chart_date=("Date", "min"),
          last_chart_date=("Date", "max"),
          entry_rank=("Rank", "first"),
          peak_rank=("Rank", "min"),
          total_weeks=("Weeks in Charts", "max"),
          top10_weeks=("Rank", lambda x: (x <= 10).sum()),
          top40_weeks=("Rank", lambda x: (x <= 40).sum()),
          avg_rank=("Rank", "mean"),
          image_url=("Image URL", "first")
      )
      .reset_index()
)

# Find the date when each song hit its peak rank
peak_dates = (
    df.loc[
        df.groupby(["Song", "Artist"])["Rank"].idxmin(),
        ["Song", "Artist", "Date"]
    ]
    .rename(columns={"Date": "peak_date"})
)

song_life = song_life.merge(
    peak_dates,
    on=["Song", "Artist"],
    how="left"
)

# Lifecycle timing metrics
song_life["lifespan_weeks"] = song_life["total_weeks"]
```

```

song_life["time_to_peak"] = (
    (song_life["peak_date"] - song_life["first_chart_date"]).dt.days / 7
)

song_life["entry_year"] = song_life["first_chart_date"].dt.year

# Flag songs whose chart run may still be incomplete near dataset endpoint
song_life["weeks_since_last_chart"] = (
    (latest_date - song_life["last_chart_date"]).dt.days / 7
)

song_life["potentially_incomplete"] = song_life["weeks_since_last_chart"] < 4

song_life.head()

```

```

Out[10]:

```

	Song	Artist	first_chart_date	last_chart_date	entry_rank	peak_rank	total_weeks	top10_weeks	top40_weeks	avg_rank
0	"1 Step Forward, 3 Steps Back"	Olivia Rodrigo	2021-06-02	2021-06-23	19	19	4	0	2	54.25000
1	"1, 2 Many"	Luke Combs & Brooks & Dunn	2019-09-25	2019-09-25	97	97	1	0	0	97.00000
2	"1, 2 Step"	Ciara Featuring Missy Elliott	2005-01-05	2005-07-20	2	2	39	12	24	19.44827
3	"1, 2, 3, 4"	Plain White T's	2009-02-04	2009-06-17	91	34	20	0	5	47.80000
4	"10,000 Hours"	Dan + Shay & Justin Bieber	2019-10-16	2020-05-06	4	4	30	14	28	15.93333

Separate COMPLETED from ALL

```

In [12]: song_life_all = song_life.copy()

song_life_completed = song_life[
    song_life["potentially_incomplete"] == False
].copy()

print("All songs:", song_life_all.shape)
print("Completed-only songs:", song_life_completed.shape)

```

All songs: (10505, 17)
Completed-only songs: (10363, 17)

Chart Analysis

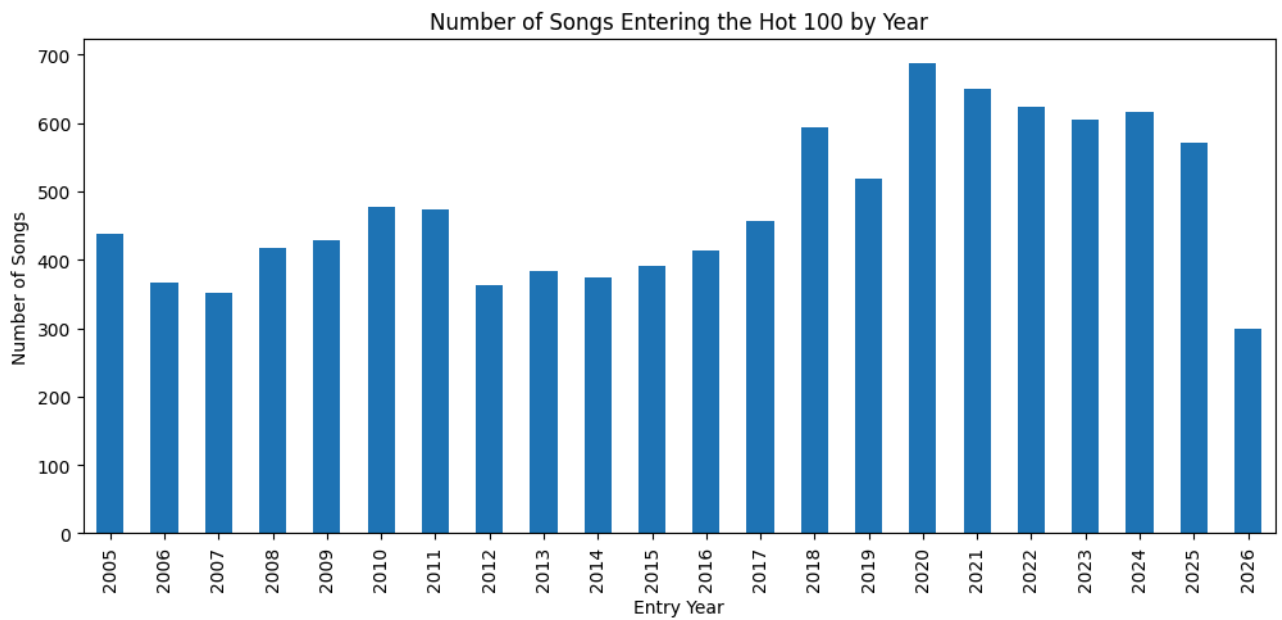
```

In [13]: # =====
# SONG COUNTS BY YEAR
# =====

song_life_all.groupby("entry_year").size().plot(
    kind="bar",
    figsize=(12, 5)
)

plt.title("Number of Songs Entering the Hot 100 by Year")
plt.xlabel("Entry Year")
plt.ylabel("Number of Songs")
plt.show()

```

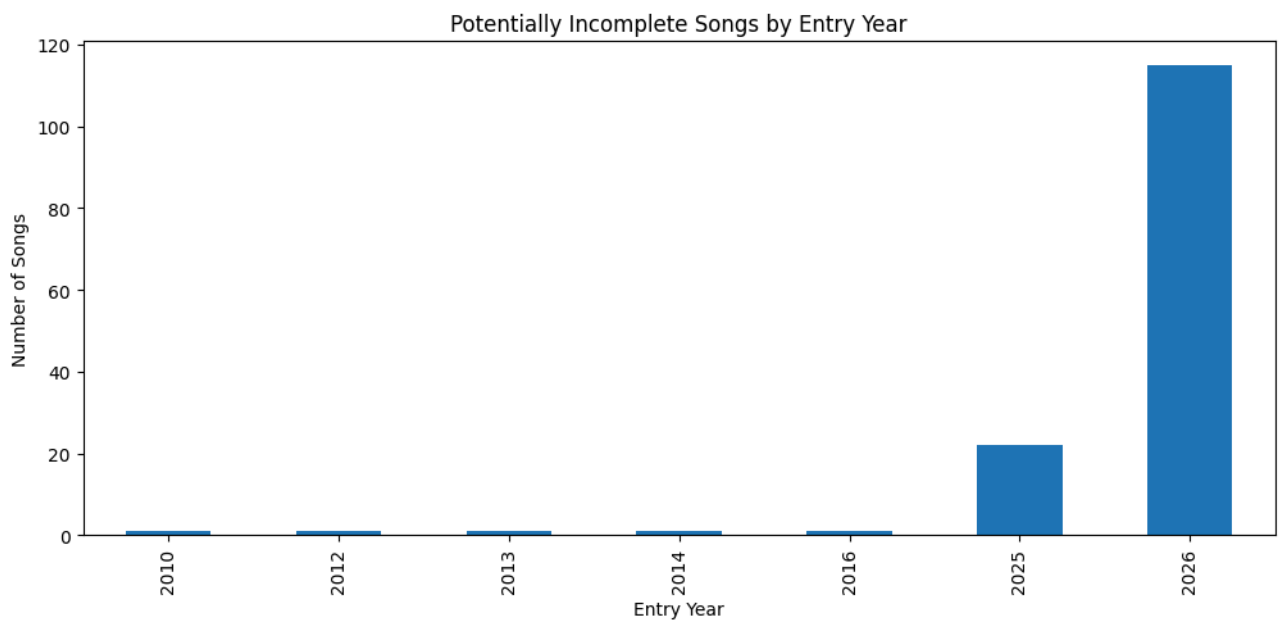


```
In [14]: # =====
# INCOMPLETE SONGS CHECK
# =====

excluded_by_year = (
    song_life[song_life["potentially_incomplete"]]
    .groupby("entry_year")
    .size()
)

excluded_by_year.plot(
    kind="bar",
    figsize=(12, 5)
)

plt.title("Potentially Incomplete Songs by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Number of Songs")
plt.show()
```



```
In [15]: # =====
# MEDIAN LIFESPAN (ALL SONGS)
# =====

median_all = (
    song_life_all
```

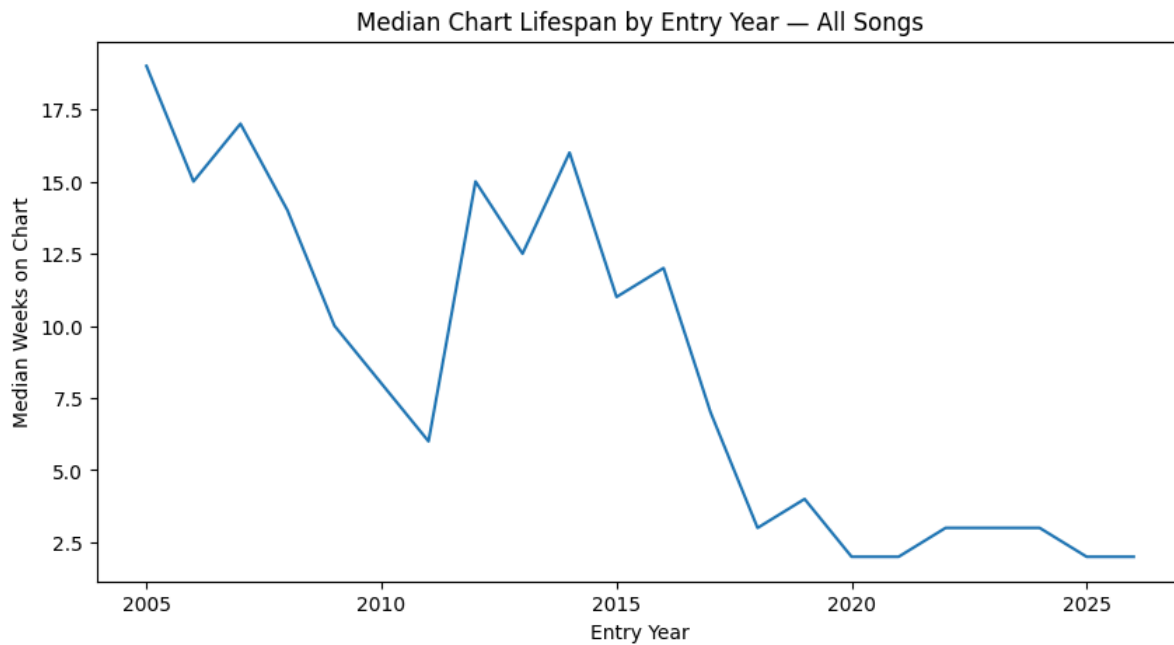
```

        .groupby("entry_year")["lifespan_weeks"]
        .median()
    )

    median_all.plot(figsize=(10, 5))

    plt.title("Median Chart Lifespan by Entry Year – All Songs")
    plt.xlabel("Entry Year")
    plt.ylabel("Median Weeks on Chart")
    plt.show()

```



```

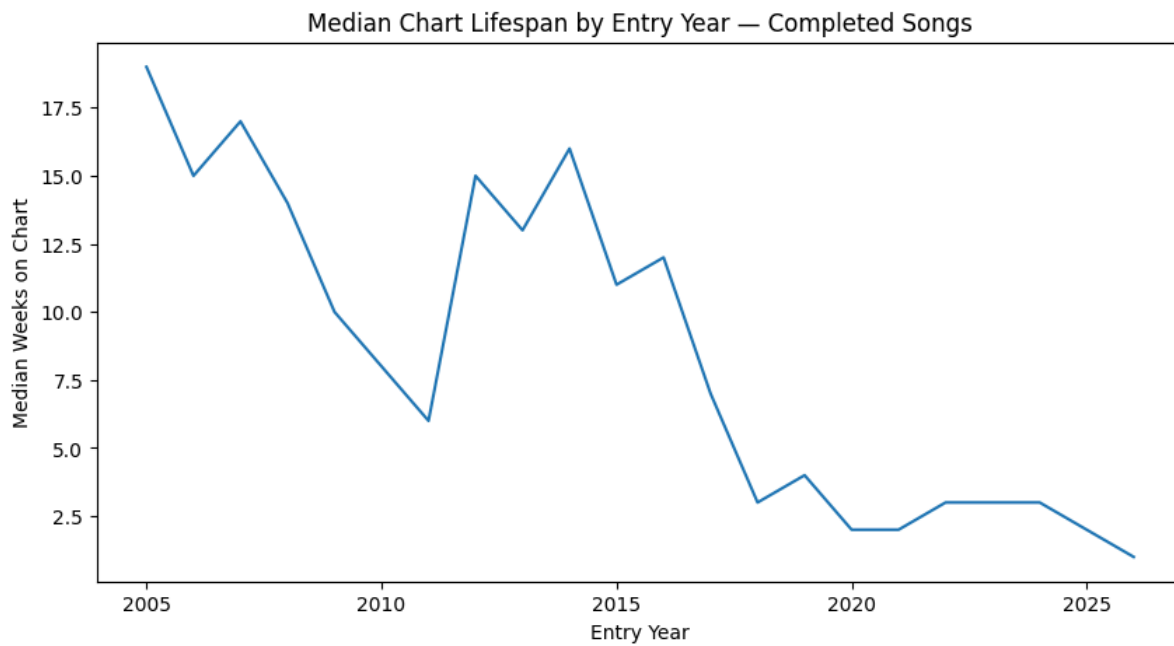
In [16]: # =====
# MEDIAN LIFESPAN (COMPLETED ONLY)
# =====

median_completed = (
    song_life_completed
    .groupby("entry_year")["lifespan_weeks"]
    .median()
)

median_completed.plot(figsize=(10, 5))

plt.title("Median Chart Lifespan by Entry Year – Completed Songs")
plt.xlabel("Entry Year")
plt.ylabel("Median Weeks on Chart")
plt.show()

```

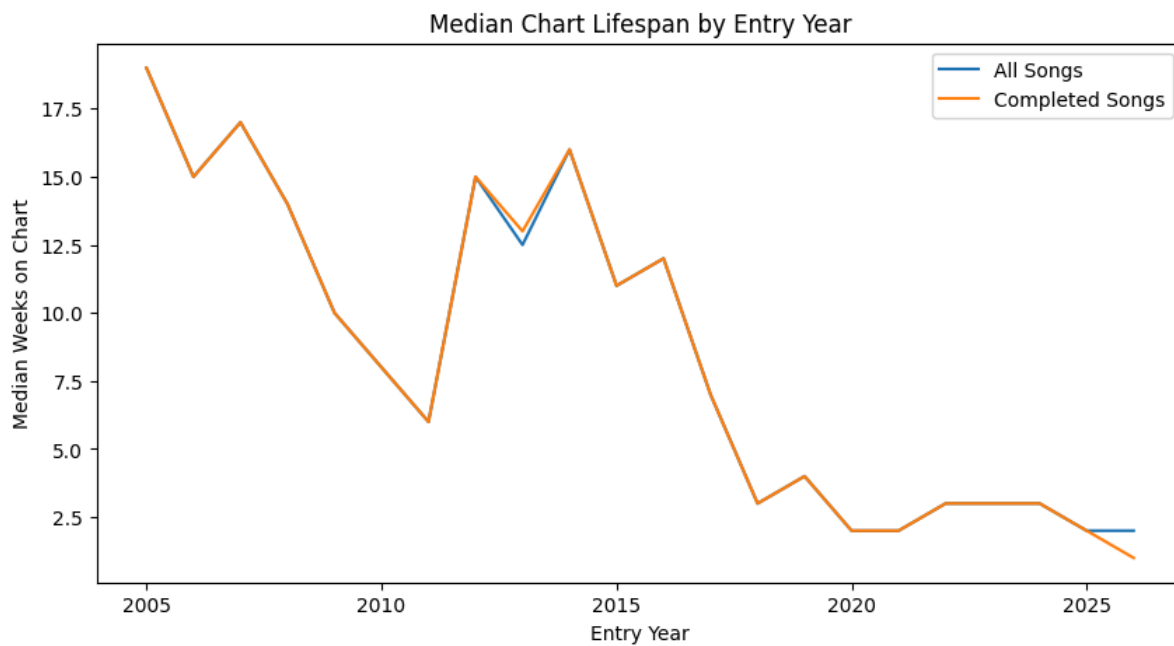


```
In [17]: # =====
# ALL VS COMPLETED COMPARISON
# =====

median_comparison = pd.DataFrame({
    "All Songs": median_all,
    "Completed Songs": median_completed
})

median_comparison.plot(figsize=(10, 5))

plt.title("Median Chart Lifespan by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Median Weeks on Chart")
plt.show()
```

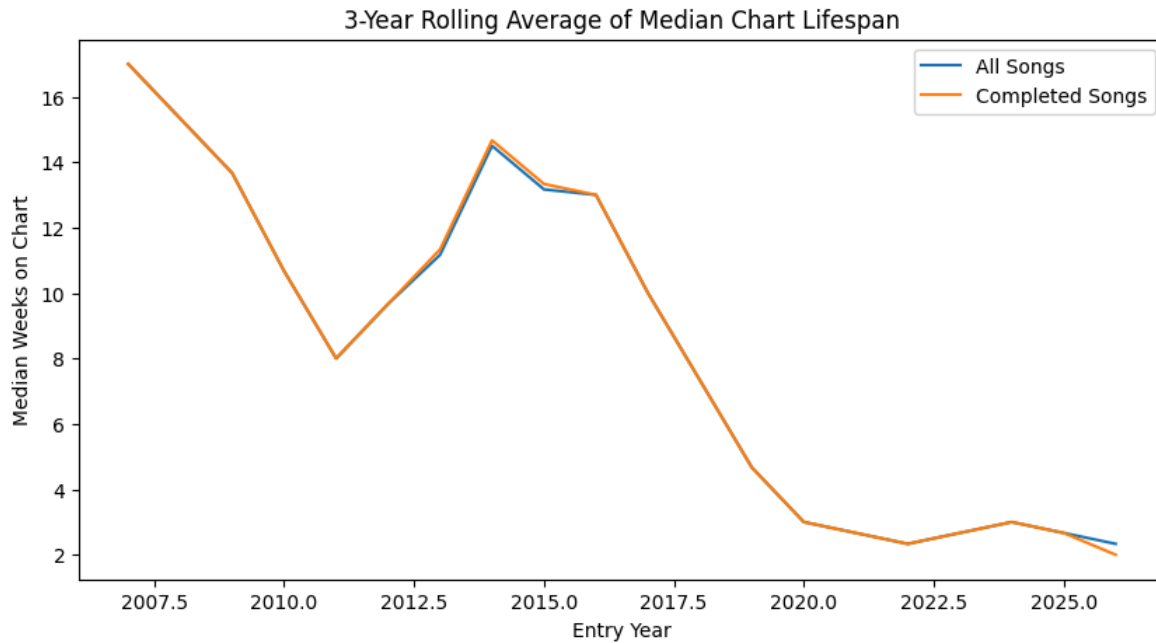


```
In [18]: # =====
# SMOOTHED TREND
# =====

median_comparison.rolling(3).mean().plot(figsize=(10, 5))

plt.title("3-Year Rolling Average of Median Chart Lifespan")
```

```
plt.xlabel("Entry Year")
plt.ylabel("Median Weeks on Chart")
plt.show()
```



```
In [19]: # =====
# MEAN LIFESPAN
# =====

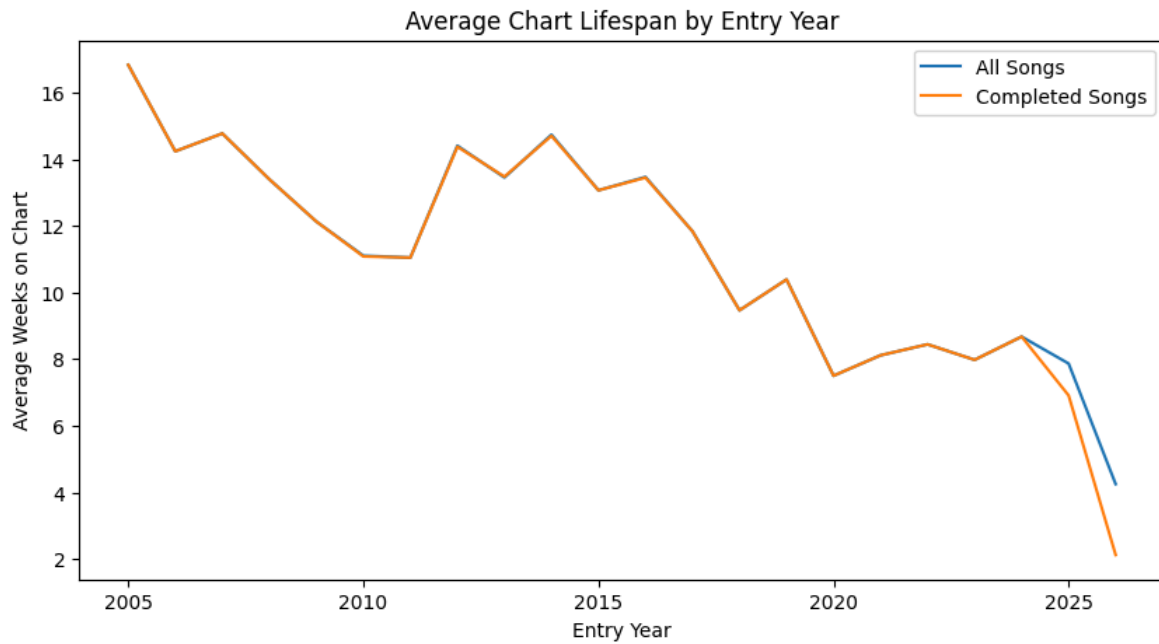
mean_all = (
    song_life_all
    .groupby("entry_year")["lifespan_weeks"]
    .mean()
)

mean_completed = (
    song_life_completed
    .groupby("entry_year")["lifespan_weeks"]
    .mean()
)

mean_comparison = pd.DataFrame({
    "All Songs": mean_all,
    "Completed Songs": mean_completed
})

mean_comparison.plot(figsize=(10, 5))

plt.title("Average Chart Lifespan by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Average Weeks on Chart")
plt.show()
```

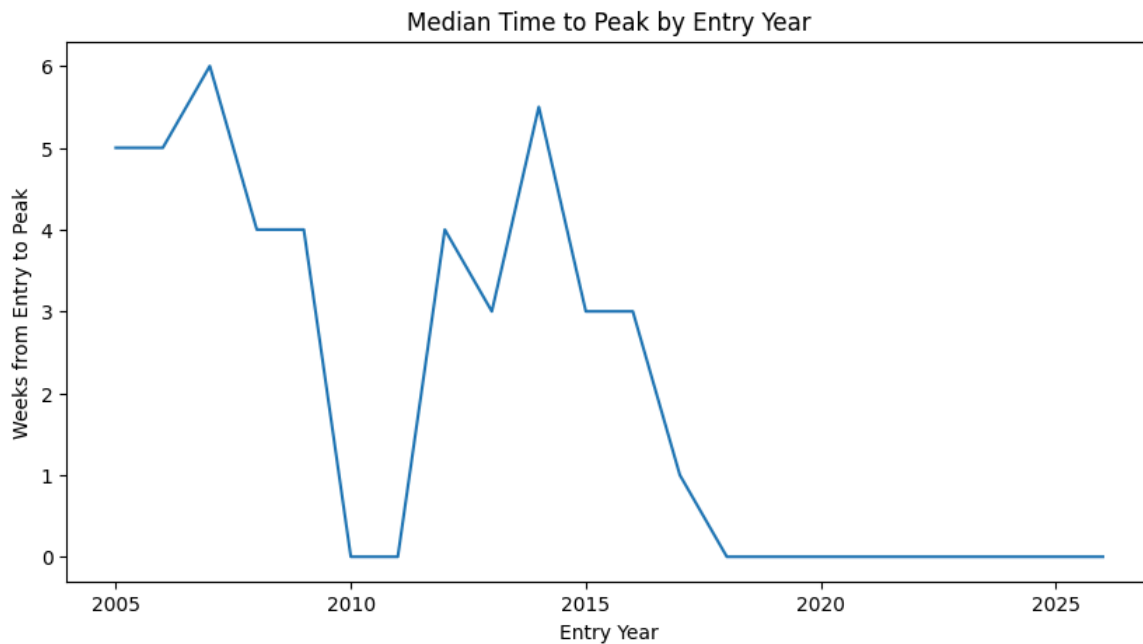


```
In [20]: # =====
# TIME TO PEAK
# =====

time_to_peak_by_year = (
    song_life_completed
    .groupby("entry_year")["time_to_peak"]
    .median()
)

time_to_peak_by_year.plot(figsize=(10, 5))

plt.title("Median Time to Peak by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Weeks from Entry to Peak")
plt.show()
```



```
In [21]: # =====
# TOP 10 PERSISTENCE
# =====

# (uses mean because median gave '0' across the board since
# less than half actually made Top 10)
```

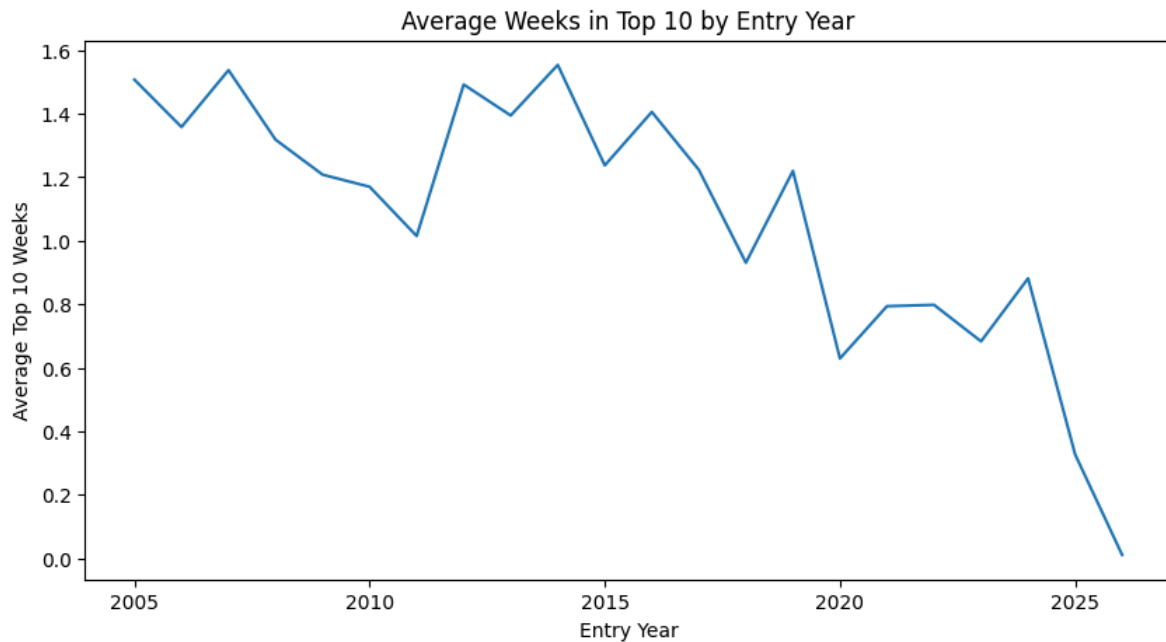
```

top10_by_year = (
    song_life_completed
    .groupby("entry_year")["top10_weeks"]
    .mean()
)

top10_by_year.plot(figsize=(10, 5))

plt.title("Average Weeks in Top 10 by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Average Top 10 Weeks")
plt.show()

```



```
In [22]: (song_life_completed["top10_weeks"] > 0).mean()
```

```
Out[22]: np.float64(0.11791952137411946)
```

```

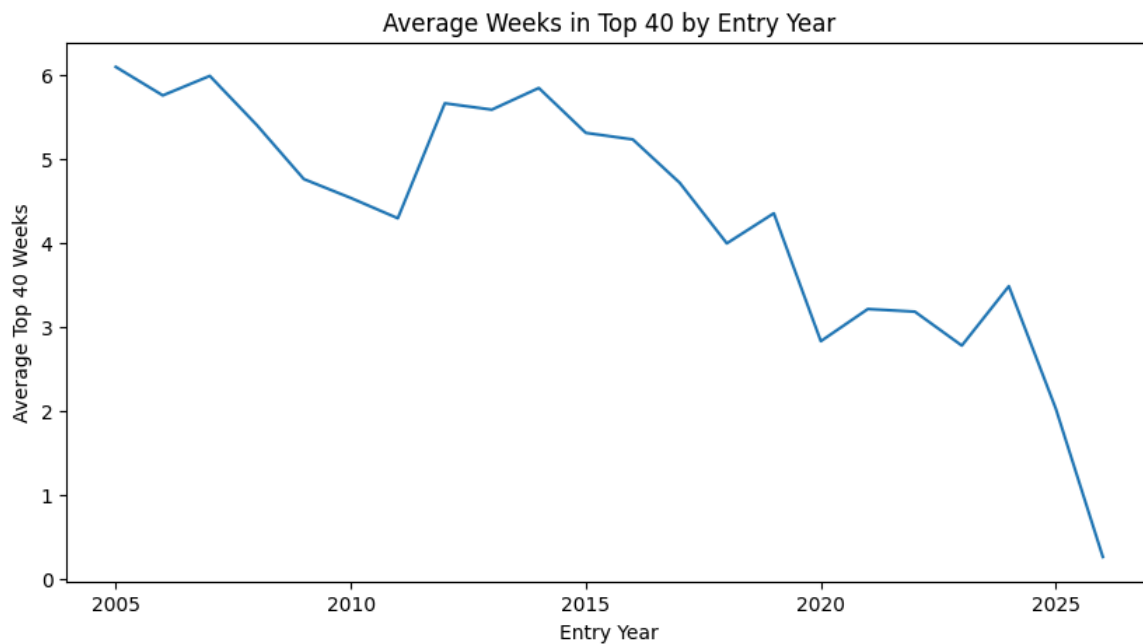
In [23]: # =====
# TOP 40 PERSISTENCE
# =====

# (uses mean because median gave '0' across the board since
# less than half actually made Top 40)
top40_by_year = (
    song_life_completed
    .groupby("entry_year")["top40_weeks"]
    .mean()
)

top40_by_year.plot(figsize=(10, 5))

plt.title("Average Weeks in Top 40 by Entry Year")
plt.xlabel("Entry Year")
plt.ylabel("Average Top 40 Weeks")
plt.show()

```

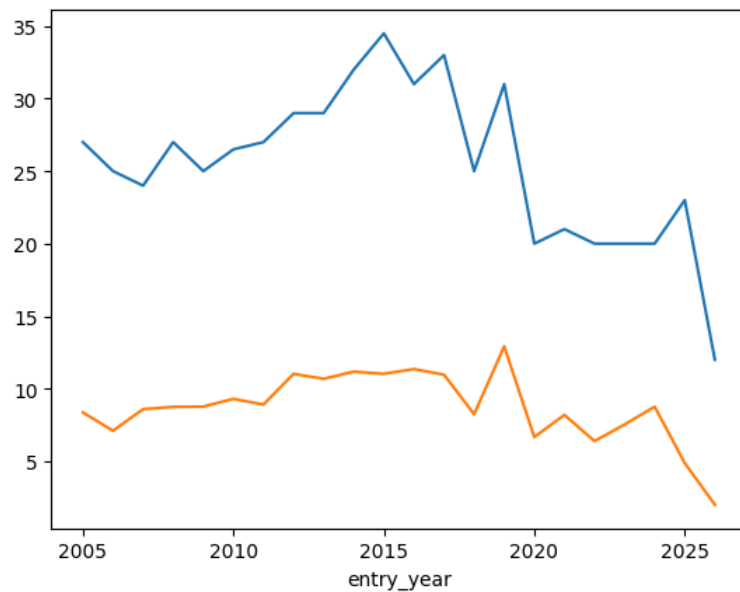



```
In [24]: # Isolating Top10 and Top40 hits
top10_hits = song_life_completed[
    song_life_completed["peak_rank"] <= 10
]

top40_hits = song_life_completed[
    song_life_completed["peak_rank"] <= 40
]

top10_hits.groupby("entry_year")["lifespan_weeks"].median().plot(), top10_hits.groupby("entry_year")["top10_weeks"]
```

Out[24]: (<Axes: xlabel='entry_year'>, <Axes: xlabel='entry_year'>)

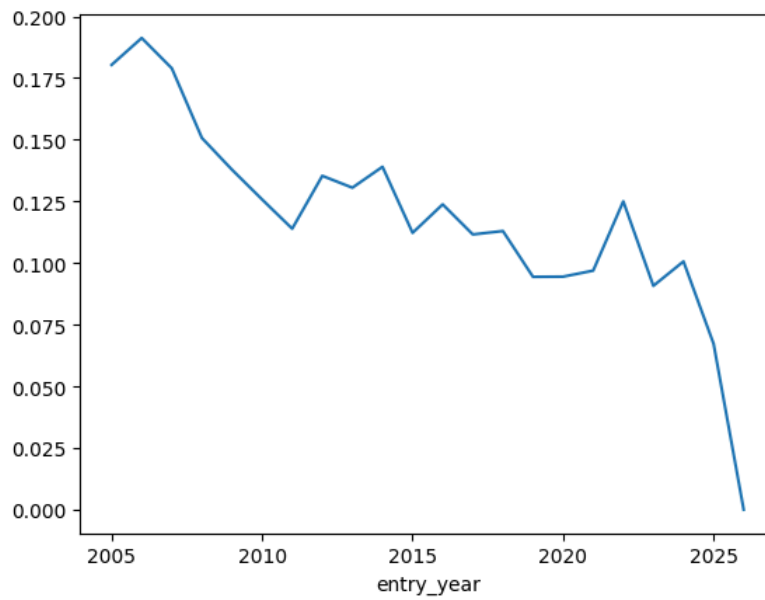


```
In [44]: # Percent of hits that reach top 10
percent_top10 = (
    song_life_completed
    .groupby("entry_year")
    .apply(lambda x: (x["peak_rank"] <= 10).mean())
)
percent_top10.plot()
```

/var/folders/bk/yngmbh1j5xdf2sk9yktp2cbr0000gn/T/ipykernel_19886/293373601.py:4: FutureWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a future version of pandas the grouping columns will be excluded from the operation. Either pass `include\_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby to silence this warning.

```
.apply(lambda x: (x["peak_rank"] <= 10).mean())
```

Out[44]: <Axes: xlabel='entry_year'>



Filtering for top 40?

```
In [141]: # unique songs by year
song_life.groupby("entry_year")["Song"].nunique().plot.bar()
```

```
Out[141]: entry_year
2005      432
2006      366
2007      349
2008      414
2009      422
2010      469
2011      460
2012      356
2013      380
2014      369
2015      387
2016      407
2017      449
2018      584
2019      511
2020      681
2021      645
2022      621
2023      604
2024      613
2025      569
2026      298
Name: Song, dtype: int64
```

```
In [142]: top40 = song_life[song_life["peak_rank"] <= 40]
top40.groupby("entry_year")["Song"].nunique()
```

```
Out[142]: entry_year
2005      214
2006      156
2007      162
2008      180
2009      177
2010      195
2011      167
2012      139
2013      158
2014      158
2015      133
2016      161
2017      146
2018      221
2019      174
2020      260
2021      233
2022      260
2023      245
2024      253
2025      179
2026       88
Name: Song, dtype: int64
```

Apple Music Preview Matching

Goal: Match Billboard songs to Apple Music tracks and retrieve 30-second preview URLs for audio feature extraction.

imports

```
In [43]: import pandas as pd
import matplotlib.pyplot as plt
import pandas as pd
import requests
import time
import re
from difflib import SequenceMatcher
```

Helper Functions

The following functions:

- clean song titles
- extract main artists
- compute similarity scores
- evaluate match quality

Pipeline

```
In [ ]: Billboard song_life dataframe (song, artist)
↓
sample songs
↓
search Apple/iTunes API
↓
get up to 10 possible results
↓
score each result
↓
pick best result
↓
label match confidence
↓
save preview URL

Output: Preview DataFrame - [search_song
search_artist
matched_song
matched_artist
collection
preview_url
```

```
match_score
title_score
artist_score
version_penalty
match_status]
```

Clean song + artist text

```
In [44]: def clean_text(text):
    text = str(text).lower()
    text = text.replace("'", "")
    text = text.replace('"', "")
    text = re.sub(r"\\..*?", "", text)
    text = re.sub(r"\\[.*?]", "", text)
    text = re.sub(r"^[^a-z0-9\\s]", " ", text)
    text = re.sub(r"\\s+", " ", text).strip()
    return text

def get_main_artist(artist):
    artist = str(artist)
    artist = re.split(
        r"featuring | feat\\. | feat | ft\\. | ft | & | and |,",
        artist,
        flags=re.IGNORECASE
    )[0]
    return artist.strip()

def similarity(a, b):
    return SequenceMatcher(None, clean_text(a), clean_text(b)).ratio()
```

Suspicious version penalty

```
In [53]: bad_version_words = [
    "piano", "acoustic", "live", "remix", "karaoke",
    "instrumental", "sped up", "slowed", "cover",
    "tribute", "demo", "reimagined", "stripped",
    "version", "tribute", "originally performed",
    "kidz bop", "16 bit", "computer game"
]

# cleaner that PRESERVES words inside parentheses/brackets
# so things like "(Acoustic)" don't disappear

def simple_clean(text):
    text = str(text).lower()
    text = text.replace("'", "")
    text = text.replace('"', "")

    text = re.sub(r"^[^a-z0-9\\s]", " ", text)
    text = re.sub(r"\\s+", " ", text).strip()

    return text

def version_penalty(track_name, collection_name):
    text = simple_clean(
        str(track_name) + " " + str(collection_name)
    )

    penalty = 0

    for word in bad_version_words:
        if word in text:
            penalty += 0.25

    return penalty
```

Score each Apple result

```
In [54]: def score_match(search_song, search_artist, result):
    matched_song = result.get("trackName", "")
    matched_artist = result.get("artistName", "")
```

```

collection = result.get("collectionName", "")
preview_url = result.get("previewUrl", None)

title_score = similarity(search_song, matched_song)
artist_score = similarity(get_main_artist(search_artist), matched_artist)
penalty = version_penalty(matched_song, collection)

preview_bonus = 0.05 if preview_url else -0.30

final_score = (
    0.65 * title_score
    + 0.30 * artist_score
    + preview_bonus
    - penalty
)

return final_score, title_score, artist_score, penalty

```

Search Apple and pick best match

```

In [114... def search_itunes_best_match(song, artist):
    """
    Search Apple/iTunes API, score possible matches, and return the best result.
    This version also records WHY a search failed.
    """

    url = "https://itunes.apple.com/search"

    # cleaner search term improves Apple results
    search_term = f"{{clean_text(song)}} {{clean_text(get_main_artist(artist))}}"

    params = {
        "term": search_term,
        "media": "music",
        "entity": "song",
        "limit": 10
    }

    try:
        response = requests.get(url, params=params, timeout=10)
    except requests.exceptions.RequestException:
        return {
            "search_song": song,
            "search_artist": artist,
            "search_term": search_term,
            "failure_reason": "REQUEST_ERROR"
        }

    if response.status_code != 200:
        return {
            "search_song": song,
            "search_artist": artist,
            "search_term": search_term,
            "failure_reason": "BAD_STATUS",
            "status_code": response.status_code
        }

    try:
        data = response.json()
    except ValueError:
        return {
            "search_song": song,
            "search_artist": artist,
            "search_term": search_term,
            "failure_reason": "JSON_ERROR",
            "response_preview": response.text[:200]
        }

    if data.get("resultCount", 0) == 0:
        return {
            "search_song": song,
            "search_artist": artist,
            "search_term": search_term,
            "failure_reason": "NO_RESULTS"
        }

    scored_results = []

    for result in data["results"]:

```

```

        score, title_score, artist_score, penalty = score_match(song, artist, result)
        scored_results.append((score, title_score, artist_score, penalty, result))

    best_score, title_score, artist_score, penalty, best_result = max(
        scored_results,
        key=lambda x: x[0]
    )

    # stricter match-status rules
    # goal: prevent same-title/wrong-artist matches and same-artist/wrong-song matches

    if (
        best_score >= 0.92
        and title_score >= 0.90
        and artist_score >= 0.65
        and penalty == 0
        and best_result.get("previewUrl") is not None
    ):
        match_status = "AUTO_ACCEPT"

    elif (
        best_score >= 0.80
        and title_score >= 0.90
        and artist_score >= 0.50
        and penalty == 0
        and best_result.get("previewUrl") is not None
    ):
        match_status = "REVIEW"

    else:
        match_status = "LOW_CONFIDENCE"

    return {
        "search_song": song,
        "search_artist": artist,
        "search_term": search_term,
        "matched_song": best_result.get("trackName"),
        "matched_artist": best_result.get("artistName"),
        "collection": best_result.get("collectionName"),
        "preview_url": best_result.get("previewUrl"),
        "match_score": best_score,
        "title_score": title_score,
        "artist_score": artist_score,
        "version_penalty": penalty,
        "match_status": match_status,
        "failure_reason": "SUCCESS"
    }

```

Artist overlap

```

In [121]: def artist_overlap(search_artist, matched_artist):
    search_clean = clean_text(search_artist)
    matched_clean = clean_text(matched_artist)

    search_words = set(search_clean.split())
    matched_words = set(matched_clean.split())

    if len(search_words) == 0 or len(matched_words) == 0:
        return 0

    overlap = search_words.intersection(matched_words)
    return len(overlap) / min(len(search_words), len(matched_words))

```

Test on 100 random songs

```

In [94]: test_songs = song_life[["Song", "Artist"]].sample(
    100,
    random_state=42
).copy()

preview_results = []

for _, row in test_songs.iterrows():
    result = search_itunes_best_match(row["Song"], row["Artist"])
    preview_results.append(result)
    time.sleep(0.5)

```

```
preview_df = pd.DataFrame(preview_results)
```

inspect results

```
In [50]: len(preview_df), len(failed)
```

```
Out[50]: (53, 47)
```

```
In [95]: preview_df["failure_reason"].value_counts()
```

```
Out[95]: failure_reason  
SUCCESS    100  
Name: count, dtype: int64
```

```
In [96]: preview_df["match_status"].value_counts(dropna=False)
```

```
Out[96]: match_status  
AUTO_ACCEPT    82  
REVIEW         11  
LOW_CONFIDENCE  7  
Name: count, dtype: int64
```

```
In [58]: preview_df.sort_values("match_score").head(20)
```

Out [58]:

	search_song	search_artist	matched_song	matched_artist	collection	preview_url	match_score	title_score
22	M3tamorphosis	Playboi Carti Featuring Kid Cudi	M3tamorphosis (8-Bit Computer Game Version)	8-Bit Arcade	By Request, Vol. 117	https://audio-ssl.itunes.apple.com/itunes-asse...	0.272000	1.000000
68	Working	Tate McRae X Khalid	she's all i wanna be	Tate McRae	she's all i wanna be - Single	https://audio-ssl.itunes.apple.com/itunes-asse...	0.356897	0.153846
65	A Boy Is A Gun	"Tyler, The Creator"	Sugar On My Tongue	Tyler, The Creator	DON'T TAP THE GLASS	https://audio-ssl.itunes.apple.com/itunes-asse...	0.470739	0.437500
83	You Got Em	Lil Durk	All My Life (feat. J. Cole)	Lil Durk	All My Life (feat. J. Cole) - Single	https://audio-ssl.itunes.apple.com/itunes-asse...	0.535714	0.285714
11	Got The Guap	Lil Uzi Vert Featuring Young Thug	Sauce It Up	Lil Uzi Vert	Luv Is Rage 2	https://audio-ssl.itunes.apple.com/itunes-asse...	0.576087	0.347826
53	Anyways	Zach Bryan	Always Willin'	Zach Bryan	With Heaven On Top	https://audio-ssl.itunes.apple.com/itunes-asse...	0.675000	0.500000
16	Twisted Transistor	Korn	Twisted Transistor	Music Box Mania	MBM Performs Korn - EP	https://audio-ssl.itunes.apple.com/itunes-asse...	0.763158	1.000000
55	Finally Found You	Enrique Iglesias Featuring Sammy Adams	Finally Found You	Morcheeba	Head Up High	https://audio-ssl.itunes.apple.com/itunes-asse...	0.772000	1.000000
26	Save Me	Jelly Roll With Lainey Wilson	Save Me (with Lainey Wilson)	Jelly Roll	Whitsitt Chapel	https://audio-ssl.itunes.apple.com/itunes-asse...	0.853846	1.000000
40	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	SAVAGE MODE II	https://audio-ssl.itunes.apple.com/itunes-asse...	0.874194	1.000000
63	Fried	¥[Math Processing Error]ign	FRIED	¥[Math Processing Error]ign	VULTURES 2	https://audio-ssl.itunes.apple.com/itunes-asse...	0.881818	1.000000
61	Girlfriend	Bow Wow & Omarion	Girlfriend	Bow Wow & Omarion	Girlfriend - Single	https://audio-ssl.itunes.apple.com/itunes-asse...	0.890909	1.000000
85	Thrift Shop	Macklemore & Ryan Lewis Featuring Wanz	Thrift Shop (feat. Wanz)	Macklemore & Ryan Lewis	The Heist	https://audio-ssl.itunes.apple.com/itunes-asse...	0.893548	1.000000
2	Otis	Jay Z Kanye West Featuring Otis Redding	Otis (feat. Otis Redding)	Kanye West & JAY-Z	Watch the Throne	https://audio-ssl.itunes.apple.com/itunes-asse...	0.893548	1.000000
41	Hummingbird	Metro Boomin & James Blake	Hummingbird	Metro Boomin & James Blake	METRO BOOMIN PRESENTS SPIDER- MAN: ACROSS THE S...	https://audio-ssl.itunes.apple.com/itunes-asse...	0.900000	1.000000
66	I Luv Her	GloRilla & T- Pain	I LUV HER	GloRilla & T- Pain	GLORIOUS	https://audio-ssl.itunes.apple.com/itunes-asse...	0.908696	1.000000
43	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale & The Big Steppers	https://audio-ssl.itunes.apple.com/itunes-asse...	0.910000	1.000000
73	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain (Deluxe Edition)	https://audio-ssl.itunes.apple.com/itunes-asse...	0.911765	1.000000
93	Verano Rosa	Karol G & Feid	Verano Rosa	KAROL G & Feid	Tropicoqueta	https://audio-ssl.itunes.apple.com/itunes-asse...	0.921053	1.000000

	search_song	search_artist	matched_song	matched_artist	collection	preview_url	match_score	title_score
44	Motive	Ariana Grande Featuring Doja Cat	motive	Ariana Grande & Doja Cat	Positions	https://audio-ssl.itunes.apple.com/itunes-asse...	0.922857	1.000000

```
In [49]: preview_df[
["search_song", "search_artist", "matched_song", "matched_artist",
"collection", "match_score", "match_status"]
].sample(20, random_state=1)
```

```
Out [49]:
```

	search_song	search_artist	matched_song	matched_artist	collection	match_score	match_status
30	Friday Night	Eric Paslay	Friday Night	Eric Paslay	Eric Paslay	1.000000	AUTO_ACCEPT
2	Otis	Jay Z Kanye West Featuring Otis Redding	Otis (feat. Otis Redding)	Kanye West & JAY-Z	Watch the Throne (Deluxe Version)	0.893548	AUTO_ACCEPT
51	X2	Lil Uzi Vert	x2	Lil Uzi Vert	Pink Tape	1.000000	AUTO_ACCEPT
32	Beg For It	Iggy Azalea Featuring M.O	Beg For It (feat. M.O)	Iggy Azalea	Reclassified	1.000000	AUTO_ACCEPT
31	I'm Still A Guy	Brad Paisley	I'm Still a Guy	Brad Paisley	5th Gear (Bonus Track Version)	1.000000	AUTO_ACCEPT
46	Electricity	Silk City x Dua Lipa	Electricity (feat. Dua Lipa) [Acoustic]	Silk City & Dua Lipa (Diplo and Marc Ronson)	Electricity (feat. Dua Lipa) [Acoustic] - Single	0.984211	AUTO_ACCEPT
34	Body Language	Jesse McCartney Featuring T-Pain	Body Language (feat. T-Pain)	Jesse McCartney	Body Language (feat. T-Pain) - Single	1.000000	AUTO_ACCEPT
39	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	SAVAGE MODE II	0.874194	AUTO_ACCEPT
45	Social Distancing	Lil Baby	Social Distancing	Lil Baby	My Turn (Deluxe)	1.000000	AUTO_ACCEPT
19	You Complete Me	Keyshia Cole	You Complete Me	Keyshia Cole	A Different Me	1.000000	AUTO_ACCEPT
10	The Matrimony	Wale Featuring Usher	The Matrimony (feat. Usher)	Wale	The Album About Nothing	1.000000	AUTO_ACCEPT
3	Paramedic!	SOB X RBE	Paramedic!	SOB X RBE	Black Panther: The Album	1.000000	AUTO_ACCEPT
21	"Heads Carolina, Tails California"	Danielle Bradbery	Heads Carolina, Tails California (The 'Voice Pe...	Danielle Bradbery	The Complete Season 4 Collection (The Voice Pe...	1.000000	AUTO_ACCEPT
49	Finesse	Drake	Finesse	Drake	Scorpion	1.000000	AUTO_ACCEPT
38	New Romantics	Taylor Swift	New Romantics (Taylor's Version)	Taylor Swift	1989 (Taylor's Version)	1.000000	AUTO_ACCEPT
41	The Boys of Fall	Kenny Chesney	The Boys of Fall	Kenny Chesney	Hemingway's Whiskey (Deluxe Edition)	1.000000	AUTO_ACCEPT
24	I Was On A Boat That Day	Old Dominion	I Was On a Boat That Day	Old Dominion	I Was On a Boat That Day - Single	1.000000	AUTO_ACCEPT
42	Motive	Ariana Grande Featuring Doja Cat	motive	Ariana Grande & Doja Cat	Positions	0.922857	AUTO_ACCEPT
40	Hummingbird	Metro Boomin & James Blake	Hummingbird	Metro Boomin & James Blake	METRO BOOMIN PRESENTS SPIDER- MAN: ACROSS THE S...	0.900000	AUTO_ACCEPT
35	Made My Hoe Faint	Future	MADE MY HOE FAINT	Future	MIXTAPE PLUTO	1.000000	AUTO_ACCEPT

```
In [60]: auto_accept = preview_df[
preview_df["match_status"] == "AUTO_ACCEPT"
]
```

```
In [63]: auto_accept[
    [
        "search_song",
        "search_artist",
        "matched_song",
        "matched_artist",
        "match_score"
    ]
].head()
```

Out[63]:

	search_song	search_artist	matched_song	matched_artist	match_score
0	"Girl, Get Up."	Doechii & SZA	girl, get up.	Doechii & SZA	0.933333
1	Bright	Echosmith	Bright	Echosmith	1.000000
3	Paramedic!	SOB X RBE	Paramedic!	SOB X RBE	1.000000
4	Let It Go	Keyshia Cole Featuring Missy Elliott & Lil Kim	Let It Go (feat. Missy Elliott & Lil 'Kim)	Keyshia Cole	1.000000
5	I Wrote The Book	Morgan Wallen	I Wrote The Book	Morgan Wallen	1.000000

```
In [71]: auto_accept.sample(20, random_state=42)
```

Out [71]:

	search_song	search_artist	matched_song	matched_artist	collection	preview_url	match_score	tit
35	Made My Hoe Faint	Future	MADE MY HOE FAINT	Future	MIXTAPE PLUTO	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
0	"Girl, Get Up."	Doechii & SZA	girl, get up.	Doechii & SZA	girl, get up. - Single	https://audio-ssl.itunes.apple.com/itunes-asse...	0.933333	
27	Just Say Dat	Gunna	just say dat	Gunna	The Last Wun	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
36	Bout A Million	Pop Smoke Featuring 21 Savage & 42 Dugg	Bout A Million (feat. 21 Savage & 42 Dugg)	Pop Smoke	Faith (Deluxe)	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
21	"Heads Carolina, Tails California"	Danielle Bradbery	Heads Carolina, Tails California (The Voice Pe...	Danielle Bradbery	The Complete Season 4 Collection (The Voice Pe...	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
33	Who	Jimin	Who	Jimin	MUSE	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
12	The Sweet Escape	Gwen Stefani Featuring Akon	The Sweet Escape (feat. Akon)	Gwen Stefani	The Sweet Escape	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
67	Wishing	"DJ Drama Featuring Chris Brown, Skeme & Lyquin"	Wishing (feat. Chris Brown, Skeme & Lyquin)	DJ Drama	Wishing (feat. Chris Brown, Skeme & Lyquin) - ...	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
5	I Wrote The Book	Morgan Wallen	I Wrote The Book	Morgan Wallen	One Thing At A Time	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
14	Like A Wrecking Ball	Eric Church	Like a Wrecking Ball	Eric Church	The Outsiders	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
59	"God, Your Mama, And Me"	Florida Georgia Line Featuring Backstreet Boys	God, Your Mama, and Me (feat. Backstreet Boys)	Florida Georgia Line	Dig Your Roots (Big Machine Radio Release Spec...	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
38	God Love Her	Toby Keith	God Love Her	Toby Keith	That Don't Make Me a Bad Guy	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
86	For A Nut	Future Featuring Gunna & Young Thug	FOR A NUT (feat. Gunna & Young Thug)	Future	I NEVER LIKED YOU	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
42	The Boys of Fall	Kenny Chesney	The Boys of Fall	Kenny Chesney	Hemingway's Whiskey (Deluxe Edition)	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
87	Rich Spirit	Kendrick Lamar	Rich Spirit	Kendrick Lamar	Mr. Morale & The Big Steppers	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
54	Happy Xmas (War Is Over)	John Legend	Happy Xmas (War Is Over)	John Legend	A Legendary Christmas	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
93	Verano Rosa	Karol G & Feid	Verano Rosa	KAROL G & Feid	Tropicoqueta	https://audio-ssl.itunes.apple.com/itunes-asse...	0.921053	
78	Yellow Flicker Beat	Lorde	Yellow Flicker Beat	Lorde	The Hunger Games: Mockingjay, Pt. 1 (Original ...	https://audio-ssl.itunes.apple.com/itunes-asse...	1.000000	
71	Double Fantasy	The Weeknd Featuring Future	Double Fantasy (feat. Future)	The Weeknd	Double Fantasy	https://audio-ssl.itunes.apple.com/itunes-	1.000000	

	search_song	search_artist	matched_song	matched_artist	collection	preview_url	match_score	tit
					(feat. Future) - Single	asse...		
48	Chill Bill	Rob toneFeaturingJ. Davi & Spooks	Chill Bill (feat. J. Davi\$ & Spooks)	Rob \$tone	Chill Bill (feat. J. Davi\$ & Spooks) - Single	https://audio- ssl.itunes.apple.com/itunes- asse...	1.000000	

Test 250 Songs

```
In [115... test_songs = song_life[["Song", "Artist"]].sample(
    250,
    random_state=42
).copy()
```

```
In [116... preview_results = []

for _, row in test_songs.iterrows():
    result = search_itunes_best_match(row["Song"], row["Artist"])
    preview_results.append(result)
    time.sleep(1.0)

preview_df = pd.DataFrame(preview_results)

preview_df["failure_reason"].value_counts()
```

```
Out[116... failure_reason
SUCCESS      250
Name: count, dtype: int64
```

```
In [124... preview_df["match_status"].value_counts(dropna=False)
```

```
Out[124... match_status
AUTO_ACCEPT      206
LOW_CONFIDENCE    24
REVIEW            20
Name: count, dtype: int64
```

```
In [125... preview_df["artist_overlap"] = preview_df.apply(
    lambda row: artist_overlap(row["search_artist"], row["matched_artist"]),
    axis=1
)

preview_df["usable_match"] = (
    (preview_df["match_status"] == "AUTO_ACCEPT")
    |
    (
        (preview_df["title_score"] >= 0.95)
        & (preview_df["artist_overlap"] >= 0.40)
        & (preview_df["version_penalty"] == 0)
        & (preview_df["preview_url"].notna())
    )
)
```

```
In [126... preview_df[
    (preview_df["usable_match"] == True)
    & (preview_df["match_status"] != "AUTO_ACCEPT")
][
    ["search_song", "search_artist", "matched_song", "matched_artist",
     "title_score", "artist_score", "artist_overlap", "version_penalty", "match_status"]
].sort_values("artist_overlap").head(30)
```

Out [126...

	search_song	search_artist	matched_song	matched_artist	title_score	artist_score	artist_overlap	versi
137	No One Mourns The Wicked	"Ariana Grande Featuring Andy Nyman, Courtney-...	No One Mourns the Wicked (feat. Andy Nyman, Co...	Wicked Movie Cast & Ariana Grande	1.0	0.590909	0.40	
2	Otis	Jay Z Kanye West Featuring Otis Redding	Otis (feat. Otis Redding)	Kanye West & JAY-Z	1.0	0.645161	0.75	
181	Now Or Never	High School Musical 3 Cast	Now or Never	The Cast of High School Musical	1.0	0.666667	0.80	
218	OK Not To Be OK	Marshmello & Demi Lovato	OK Not To Be OK	Marshmello & Demi Lovato	1.0	0.625000	1.00	
216	Sharks	"Lil Wayne, Jelly Roll & Big Sean"	Sharks	Lil Wayne, Jelly Roll & Big Sean	1.0	0.473684	1.00	
203	Fine China	Future & Juice WRLD	Fine China	Future & Juice WRLD	1.0	0.521739	1.00	
164	Amen	Shaboozey & Jelly Roll	Amen	Shaboozey & Jelly Roll	1.0	0.620690	1.00	
125	Hate Me	Ellie Goulding & Juice WRLD	Hate Me	Ellie Goulding & Juice WRLD	1.0	0.717949	1.00	
110	Santa Baby	Eartha Kitt With Henri Rene And His Orchestra	Santa Baby (with Henri René and His Orchestra)	Eartha Kitt	1.0	0.578947	1.00	
219	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	1.0	0.571429	1.00	
106	Wild Child	Kenny Chesney With Grace Potter	Wild Child (with Grace Potter)	Kenny Chesney	1.0	0.590909	1.00	
73	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain	Selena Gomez & The Scene	1.0	0.705882	1.00	
66	I Luv Her	GloRilla & T-Pain	I LUV HER	GloRilla & T-Pain	1.0	0.695652	1.00	
63	Fried	¥ : Kanye West & Ty Dolla ign	FRIED	¥ , Kanye West & Ty Dolla ign	1.0	0.606061	1.00	
61	Girlfriend	Bow Wow & Omarion	Girlfriend	Bow Wow & Omarion	1.0	0.636364	1.00	
43	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale	Kendrick Lamar & Tanna Leone	1.0	0.700000	1.00	
41	Hummingbird	Metro Boomin & James Blake	Hummingbird	Metro Boomin & James Blake	1.0	0.666667	1.00	
40	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	1.0	0.580645	1.00	
26	Save Me	Jelly Roll With Lainey Wilson	Save Me (with Lainey Wilson)	Jelly Roll	1.0	0.512821	1.00	
85	Thrift Shop	Macklemore & Ryan Lewis Featuring Wanz	Thrift Shop (feat. Wanz)	Macklemore & Ryan Lewis	1.0	0.645161	1.00	
240	Made For Now	Janet & Daddy Yankee	Made For Now	Janet Jackson & Daddy Yankee	1.0	0.322581	1.00	

In [127...

```
auto_accept_count = (  
    preview_df["match_status"] == "AUTO_ACCEPT"  
).sum()  
  
usable_count = preview_df["usable_match"].sum()  
  
print("AUTO_ACCEPT:", auto_accept_count)  
print("USABLE:", usable_count)  
print("Recovered:", usable_count - auto_accept_count)
```

AUTO_ACCEPT: 206
USABLE: 227
Recovered: 21

In [118...

```
preview_df[  
    preview_df["match_status"] != "AUTO_ACCEPT"  
][  
    ["search_song", "search_artist", "matched_song", "matched_artist",
```

```
    "collection", "match_score", "title_score", "artist_score", "version_penalty", "match_status"]  
].sort_values("match_score").head(30)
```

Out [118...

	search_song	search_artist	matched_song	matched_artist	collection	match_score	title_score	artist_score	version_pen
22	M3tamorphosis	Playboi Carti Featuring Kid Cudi	M3tamorphosis (8-Bit Computer Game Version)	8-Bit Arcade	By Request, Vol. 117	0.272000	1.000000	0.240000	
68	Working	Tate McRae X Khalid	she's all i wanna be	Tate McRae	she's all i wanna be - Single	0.356897	0.153846	0.689655	
153	Dis 1 Got It	Playboi Carti	Let It Go (Frozen)	Lullaby Baby Trio	Disney Lullabies Classic Renditions of Disney ...	0.397619	0.380952	0.333333	
83	You Got Em	Lil Durk	All My Life (feat. J. Cole)	Lil Durk	All My Life (feat. J. Cole) - Single	0.535714	0.285714	1.000000	
226	F**k The Industry Pt. 2	YoungBoy Never Broke Again	Make No Sense	YoungBoy Never Broke Again	AI YoungBoy 2	0.541176	0.294118	1.000000	
186	Untouchable	YoungBoy Never Broke Again	Slime Mentality	YoungBoy Never Broke Again	AI YoungBoy 2	0.550000	0.307692	1.000000	
193	Outfit	Lil Baby & 21 Savage	Close Friends	Lil Baby	Drip Harder	0.555263	0.315789	1.000000	
11	Got The Guap	Lil Uzi Vert Featuring Young Thug	Sauce It Up	Lil Uzi Vert	Luv Is Rage 2	0.576087	0.347826	1.000000	
182	Cold Heart (PNAU Remix)	Elton John & Dua Lipa	Cold Heart (PNAU Remix)	Elton John & Dua Lipa	The Lockdown Sessions	0.656897	1.000000	0.689655	
53	Anyways	Zach Bryan	Always Willin'	Zach Bryan	With Heaven On Top	0.675000	0.500000	1.000000	
149	C'mon	Ke\$ha	C'mon (Cutmore Radio Mix)	Kesha	C'mon (Remixes) - EP	0.690000	1.000000	0.800000	
238	Blood On The Leaves	Kanye West	Blood on the Leaves	Madh	Sayonara - EP	0.742857	1.000000	0.142857	
183	Live Your Life	T.I. Featuring Rihanna	Live Your Life (feat. Rihanna)	T.I.	Paper Trail	0.750000	1.000000	1.000000	
112	Long Live (Taylor's Version)	Taylor Swift	Long Live	Taylor Swift	Speak Now	0.750000	1.000000	1.000000	
118	Love This Life	T.I.	Love This Life	T.I.	Trouble Man: Heavy is the Head (Deluxe Version)	0.750000	1.000000	1.000000	
16	Twisted Transistor	Korn	Twisted Transistor	Music Box Mania	MBM Performs Korn - EP	0.763158	1.000000	0.210526	
65	A Boy Is A Gun	"Tyler, The Creator"	A BOY IS A GUN	Vivid The Producer	A BOY IS A GUN - Single	0.778261	1.000000	0.260870	
240	Made For Now	Janet & Daddy Yankee	Made For Now	Janet Jackson & Daddy Yankee	Made For Now - Single	0.796774	1.000000	0.322581	
140	Better Man (Taylor's Version) (From The Vault)	Taylor Swift	Better Man	Leon Bridges	Coming Home (Deluxe)	0.800000	1.000000	0.333333	
197	abcdefu	GAYLE	Abcdefu	Madiilyn	Abcdefu - Single	0.800000	1.000000	0.333333	

	search_song	search_artist	matched_song	matched_artist	collection	match_score	title_score	artist_score	version_pen
55	Finally Found You	Enrique Iglesias Featuring Sammy Adams	Finally Found You	EDGAR SIRANTANG	Finally Found You - Single	0.816129	1.000000	0.387097	
163	Sorry Not Sorry	"Tyler, The Creator"	Sorry Not Sorry	Bryson Tiller	T R A P S O U L	0.833333	1.000000	0.444444	
216	Sharks	"Lil Wayne, Jelly Roll & Big Sean"	Sharks	Lil Wayne, Jelly Roll & Big Sean	Tha Carter VI	0.842105	1.000000	0.473684	
26	Save Me	Jelly Roll With Lainey Wilson	Save Me (with Lainey Wilson)	Jelly Roll	Whitsitt Chapel	0.853846	1.000000	0.512821	
203	Fine China	Future & Juice WRLD	Fine China	Future & Juice WRLD	Fine China - Single	0.856522	1.000000	0.521739	
219	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	Fast & Furious 6 (Original Motion Picture Soun...	0.871429	1.000000	0.571429	
110	Santa Baby	Eartha Kitt With Henri Rene And His Orchestra	Santa Baby (with Henri René and His Orchestra)	Eartha Kitt	Santa Baby	0.873684	1.000000	0.578947	
40	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	SAVAGE MODE II	0.874194	1.000000	0.580645	
106	Wild Child	Kenny Chesney With Grace Potter	Wild Child (with Grace Potter)	Kenny Chesney	The Big Revival	0.877273	1.000000	0.590909	
137	No One Mourns The Wicked	"Ariana Grande Featuring Andy Nyman, Courtney-...	No One Mourns the Wicked (feat. Andy Nyman, Co...	Wicked Movie Cast & Ariana Grande	Wicked: The Soundtrack	0.877273	1.000000	0.590909	

```
In [119... preview_df[
    preview_df["match_status"] == "AUTO_ACCEPT"
][
    [
        "search_song",
        "search_artist",
        "matched_song",
        "matched_artist"
    ]
].sample(50, random_state=1)
```


	search_song	search_artist	matched_song	matched_artist
96	I Smile	Kirk Franklin	I Smile	Kirk Franklin
119	Come Away To The Water	Maroon 5 Featuring Rozzi Crane	Come Away to the Water (feat. Rozzi Crane)	Maroon 5
185	Flocky Flocky	Don Toliver Featuring Travis Scott	Flocky Flocky (feat. Travis Scott)	Don Toliver
52	Electricity	Silk City x Dua Lipa	Electricity (feat. Diplo & Mark Ronson)	Silk City, Dua Lipa
48	Chill Bill	Rob <i>tone</i> FeaturingJ. Davi & Spooks	Chill Bill (feat. J. Davi\$ & Spooks)	Rob \$tone
84	Keep Your Head Up	Andy Grammer	Keep Your Head Up	Andy Grammer
120	Defying Gravity	Cynthia Erivo Featuring Ariana Grande	Defying Gravity (feat. Ariana Grande)	Cynthia Erivo
21	"Heads Carolina, Tails California"	Danielle Bradbery	Heads Carolina, Tails California (The Voice Pe...	Danielle Bradbery
187	Evacuate The Dancefloor	Cascada	Evacuate the Dancefloor	Cascada
78	Yellow Flicker Beat	Lorde	Yellow Flicker Beat	Lorde
5	I Wrote The Book	Morgan Wallen	I Wrote The Book	Morgan Wallen
211	Already Gone	Sugarland	Already Gone	Sugarland
175	WE (Warm Embrace)	Chris Brown	WE (Warm Embrace)	Chris Brown
46	Single Saturday Night	Cole Swindell	Single Saturday Night	Cole Swindell
34	Body Language	Jesse McCartney Featuring T-Pain	Body Language (feat. T-Pain)	Jesse McCartney
209	Graveyard	Halsey	Graveyard	Halsey
233	When I'm Gone	Wiz Khalifa	When I'm Gone	Wiz Khalifa
38	God Love Her	Toby Keith	God Love Her	Toby Keith
221	Half Of Me	Thomas Rhett Featuring Riley Green	Half Of Me (feat. Riley Green)	Thomas Rhett
232	Carla's Song	Harry Styles	Carla's Song	Harry Styles
212	Total Eclipse Of The Heart	Glee Cast Featuring Jonathan Groff	Total Eclipse Of The Heart (feat. Jonathan Groff)	Glee Cast
36	Bout A Million	Pop Smoke Featuring 21 Savage & 42 Dugg	Bout A Million (feat. 21 Savage & 42 Dugg)	Pop Smoke
13	Levitate	Hadouken!	Levitate	Hadouken!
144	That Way	Wale Featuring Jeremih & Rick Ross	That Way (feat. Jeremih & Rick Ross)	Wale
228	Gemstone	Don Toliver	Gemstone	Don Toliver
39	New Romantics	Taylor Swift	New Romantics	Taylor Swift
102	These Words	Natasha Bedingfield	These Words	Natasha Bedingfield
227	Modern Day	Migos	Modern Day	Migos
145	Show Me	Kid Ink Featuring Chris Brown	Show Me (feat. Chris)	Kid Ink
42	The Boys of Fall	Kenny Chesney	The Boys of Fall	Kenny Chesney
180	Double Down	Lil Baby	Double Down	Lil Baby
33	Who	Jimin	Who	Jimin
19	You Complete Me	Keyshia Cole	You Complete Me	Keyshia Cole
157	Jump Then Fall	Taylor Swift	Jump Then Fall	Taylor Swift
141	Rude Boy	Rihanna	Rude Boy	Rihanna
224	Temptation	Future	Temptation	Future
32	Beg For It	Iggy Azalea Featuring M.O	Beg For It (feat. M.O)	Iggy Azalea
57	Ten Feet Tall	Afrojack Featuring Wrabel	Ten Feet Tall (feat. Wrabel)	AFROJACK
74	All My Life	Lil Durk Featuring J. Cole	All My Life (feat. J. Cole)	Lil Durk
249	Last Friday Night	Glee Cast	Last Friday Night (feat. Darren Criss)	Glee Cast
231	Since Way Back	Drake Featuring PARTYNEXTDOOR	Since Way Back (feat. PARTYNEXTDOOR)	Drake
129	Time Warp	Glee Cast	Time Warp	Glee Cast

	search_song	search_artist	matched_song	matched_artist
109	Northern Attitude	Noah Kahan With Hozier	Northern Attitude	Noah Kahan & Hozier
178	4KT Baby	YoungBoy Never Broke Again	4KT Baby	YoungBoy Never Broke Again
173	Universal Mind Control	Common Featuring Pharrell	Universal Mind Control (UMC)	Common
116	Visiting Hours	Ed Sheeran	Visiting Hours	Ed Sheeran
91	Cuando Me Enamoro	Enrique Iglesias Featuring Juan Luis Guerra	Cuando Me Enamoro (feat. Juan Luis Guerra)	Enrique Iglesias
62	X2	Lil Uzi Vert	x2	Lil Uzi Vert
142	Spin Again	Lil Uzi Vert	Spin Again	Lil Uzi Vert
126	Goodnight N Go	Ariana Grande	goodnight n go	Ariana Grande

```
In [120... review_df = preview_df[
    preview_df["match_status"] != "AUTO_ACCEPT"
].copy()

review_df[
    [
        "search_song",
        "search_artist",
        "matched_song",
        "matched_artist",
        "title_score",
        "artist_score",
        "match_status"
    ]
].head(50)
```

	search_song	search_artist	matched_song	matched_artist	title_score	artist_score	match_status
2	Otis	Jay Z Kanye West Featuring Otis Redding	Otis (feat. Otis Redding)	Kanye West & JAY-Z	1.000000	0.645161	REVIEW
11	Got The Guap	Lil Uzi Vert Featuring Young Thug	Sauce It Up	Lil Uzi Vert	0.347826	1.000000	LOW_CONFIDENCE
16	Twisted Transistor	Korn	Twisted Transistor	Music Box Mania	1.000000	0.210526	LOW_CONFIDENCE
22	M3tamorphosis	Playboi Carti Featuring Kid Cudi	M3tamorphosis (8-Bit Computer Game Version)	8-Bit Arcade	1.000000	0.240000	LOW_CONFIDENCE
26	Save Me	Jelly Roll With Lainey Wilson	Save Me (with Lainey Wilson)	Jelly Roll	1.000000	0.512821	REVIEW
40	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	1.000000	0.580645	REVIEW
41	Hummingbird	Metro Boomin & James Blake	Hummingbird	Metro Boomin & James Blake	1.000000	0.666667	REVIEW
43	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale	Kendrick Lamar & Tanna Leone	1.000000	0.700000	REVIEW
53	Anyways	Zach Bryan	Always Willin'	Zach Bryan	0.500000	1.000000	LOW_CONFIDENCE
55	Finally Found You	Enrique Iglesias Featuring Sammy Adams	Finally Found You	EDGAR SIRANTANG	1.000000	0.387097	LOW_CONFIDENCE
61	Girlfriend	Bow Wow & Omarion	Girlfriend	Bow Wow & Omarion	1.000000	0.636364	REVIEW
63	Fried	¥ : Kanye West & Ty Dolla ign	FRIED	¥ , Kanye West & Ty Dolla ign	1.000000	0.606061	REVIEW
65	A Boy Is A Gun	"Tyler, The Creator"	A BOY IS A GUN	Vivid The Producer	1.000000	0.260870	LOW_CONFIDENCE
66	I Luv Her	GloRilla & T-Pain	I LUV HER	GloRilla & T-Pain	1.000000	0.695652	REVIEW
68	Working	Tate McRae X Khalid	she's all i wanna be	Tate McRae	0.153846	0.689655	LOW_CONFIDENCE
73	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain	Selena Gomez & The Scene	1.000000	0.705882	REVIEW
83	You Got Em	Lil Durk	All My Life (feat. J. Cole)	Lil Durk	0.285714	1.000000	LOW_CONFIDENCE
85	Thrift Shop	Macklemore & Ryan Lewis Featuring Wanz	Thrift Shop (feat. Wanz)	Macklemore & Ryan Lewis	1.000000	0.645161	REVIEW
106	Wild Child	Kenny Chesney With Grace Potter	Wild Child (with Grace Potter)	Kenny Chesney	1.000000	0.590909	REVIEW
110	Santa Baby	Eartha Kitt With Henri Rene And His Orchestra	Santa Baby (with Henri René and His Orchestra)	Eartha Kitt	1.000000	0.578947	REVIEW
112	Long Live (Taylor's Version)	Taylor Swift	Long Live	Taylor Swift	1.000000	1.000000	LOW_CONFIDENCE
118	Love This Life	T.I.	Love This Life	T.I.	1.000000	1.000000	LOW_CONFIDENCE
125	Hate Me	Ellie Goulding & Juice WRLD	Hate Me	Ellie Goulding & Juice WRLD	1.000000	0.717949	REVIEW
137	No One Mourns The Wicked	"Ariana Grande Featuring Andy Nyman, Courtney-...	No One Mourns the Wicked (feat. Andy Nyman, Co...	Wicked Movie Cast & Ariana Grande	1.000000	0.590909	REVIEW
140	Better Man (Taylor's Version) (From The Vault)	Taylor Swift	Better Man	Leon Bridges	1.000000	0.333333	LOW_CONFIDENCE
149	C'mon	Ke\$ha	C'mon (Cutmore Radio Mix)	Kesha	1.000000	0.800000	LOW_CONFIDENCE
153	Dis 1 Got It	Playboi Carti	Let It Go (Frozen)	Lullaby Baby Trio	0.380952	0.333333	LOW_CONFIDENCE

	search_song	search_artist	matched_song	matched_artist	title_score	artist_score	match_status
163	Sorry Not Sorry	"Tyler, The Creator"	Sorry Not Sorry	Bryson Tiller	1.000000	0.444444	LOW_CONFIDENCE
164	Amen	Shaboozey & Jelly Roll	Amen	Shaboozey & Jelly Roll	1.000000	0.620690	REVIEW
172	Up To The Mountain	Kelly Clarkson Featuring Jeff Beck	Up to the Mountain	Emily Wilson	1.000000	0.615385	REVIEW
181	Now Or Never	High School Musical 3 Cast	Now or Never	The Cast of High School Musical	1.000000	0.666667	REVIEW
182	Cold Heart (PNAU Remix)	Elton John & Dua Lipa	Cold Heart (PNAU Remix)	Elton John & Dua Lipa	1.000000	0.689655	LOW_CONFIDENCE
183	Live Your Life	T.I. Featuring Rihanna	Live Your Life (feat. Rihanna)	T.I.	1.000000	1.000000	LOW_CONFIDENCE
186	Untouchable	YoungBoy Never Broke Again	Slime Mentality	YoungBoy Never Broke Again	0.307692	1.000000	LOW_CONFIDENCE
193	Outfit	Lil Baby & 21 Savage	Close Friends	Lil Baby	0.315789	1.000000	LOW_CONFIDENCE
197	abcdefu	GAYLE	Abcdefu	Madilyn	1.000000	0.333333	LOW_CONFIDENCE
203	Fine China	Future & Juice WRLD	Fine China	Future & Juice WRLD	1.000000	0.521739	REVIEW
206	Fuck The World	Rod Wave	F**k the World	Rod Wave	0.888889	1.000000	LOW_CONFIDENCE
216	Sharks	"Lil Wayne, Jelly Roll & Big Sean"	Sharks	Lil Wayne, Jelly Roll & Big Sean	1.000000	0.473684	LOW_CONFIDENCE
218	OK Not To Be OK	Marshmello & Demi Lovato	OK Not To Be OK	Marshmello & Demi Lovato	1.000000	0.625000	REVIEW
219	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	We Own It (Fast & Furious)	2 Chainz & Wiz Khalifa	1.000000	0.571429	REVIEW
226	F**k The Industry Pt. 2	YoungBoy Never Broke Again	Make No Sense	YoungBoy Never Broke Again	0.294118	1.000000	LOW_CONFIDENCE
238	Blood On The Leaves	Kanye West	Blood on the Leaves	Madh	1.000000	0.142857	LOW_CONFIDENCE
240	Made For Now	Janet & Daddy Yankee	Made For Now	Janet Jackson & Daddy Yankee	1.000000	0.322581	LOW_CONFIDENCE

Test 250 more

```
In [128... test_songs_2 = song_life[["Song", "Artist"]].sample(
    250,
    random_state=99
).copy()

preview_results_2 = []

for _, row in test_songs_2.iterrows():
    result = search_itunes_best_match(row["Song"], row["Artist"])
    preview_results_2.append(result)
    time.sleep(1.0)

preview_df_2 = pd.DataFrame(preview_results_2)
```

```
In [129... preview_df_2["artist_overlap"] = preview_df_2.apply(
    lambda row: artist_overlap(row["search_artist"], row["matched_artist"]),
    axis=1
)

preview_df_2["usable_match"] = (
    (preview_df_2["match_status"] == "AUTO_ACCEPT")
    |
    (
        (preview_df_2["title_score"] >= 0.95)
        & (preview_df_2["artist_overlap"] >= 0.40)
        & (preview_df_2["version_penalty"] == 0)
        & (preview_df_2["preview_url"].notna())
    )
)
```

```
In [130... preview_df_2["failure_reason"].value_counts()
```

```
Out[130...] failure_reason
SUCCESS      240
BAD_STATUS    9
NO_RESULTS    1
Name: count, dtype: int64
```

```
In [131...] preview_df_2["match_status"].value_counts(dropna=False)
```

```
Out[131...] match_status
AUTO_ACCEPT      193
LOW_CONFIDENCE    30
REVIEW            17
NaN               10
Name: count, dtype: int64
```

```
In [132...] preview_df_2["usable_match"].value_counts()
```

```
Out[132...] usable_match
True      214
False     36
Name: count, dtype: int64
```

```
In [133...] preview_df_2[
    (preview_df_2["usable_match"] == True)
    & (preview_df_2["match_status"] != "AUTO_ACCEPT")
][
    ["search_song", "search_artist", "matched_song", "matched_artist",
     "title_score", "artist_overlap", "version_penalty", "match_status"]
].head(30)
```

Out [133...

	search_song	search_artist	matched_song	matched_artist	title_score	artist_overlap	version_penalty	
6	The Adventures Of Moon Man & Slim Shady	Kid Cudi & Eminem	The Adventures of Moon Man & Slim Shady	Kid Cudi & Eminem	1.0	1.0	0.0	
36	Sunflower (Spider-Man: Into The Spider-Verse)	Post Malone & Swae Lee	Sunflower (Spider-Man: Into the Spider-Verse)	Post Malone & Swae Lee	1.0	1.0	0.0	
38	For You (Fifty Shades Freed)	Liam Payne & Rita Ora	For You (Fifty Shades Freed)	Liam Payne & Rita Ora	1.0	1.0	0.0	
48	Rumors	Lizzo Featuring Cardi B	Rumors	Lizzo & Cardi B	1.0	1.0	0.0	
50	LV Bag	"Don Toliver, J-Hope, Pharrell Williams & Speedy"	LV Bag (feat. Pharrell Williams)	Don Toliver, Speedy & j-hope	1.0	1.0	0.0	
55	HVN On Earth	Lil Tecca & Kodak Black	HVN ON EARTH	Lil Tecca & Kodak Black	1.0	1.0	0.0	
67	Potion	"Calvin Harris, Dua Lipa & Young Thug"	Potion (Mixed)	Calvin Harris, Young Thug & Dua Lipa	1.0	1.0	0.0	
70	OK Not To Be OK	Marshmello & Demi Lovato	OK Not To Be OK	Marshmello & Demi Lovato	1.0	1.0	0.0	
85	Bandit	Juice WRLD & YoungBoy Never Broke Again	Bandit	Juice WRLD & YoungBoy Never Broke Again	1.0	1.0	0.0	L
88	One Right Now	Post Malone & The Weeknd	One Right Now	Post Malone & The Weeknd	1.0	1.0	0.0	
103	First Time	Kygo & Ellie Goulding	First Time	Kygo & Ellie Goulding	1.0	1.0	0.0	L
116	Rewrite The Stars	Zac Efron & Zendaya	Rewrite the Stars	Zac Efron & Zendaya	1.0	1.0	0.0	
119	Reflections Laughing	"The Weeknd, Travis Scott & Florence + The Mac...	Reflections Laughing	The Weeknd, Travis Scott & Florence + the Machine	1.0	1.0	0.0	L
149	Luv Bad Bitches	"Future, Metro Boomin & Brownstone"	Luv Bad Bitches	Future, Metro Boomin & Brownstone	1.0	1.0	0.0	L
156	American Honey	Lady Antebellum	American Honey	Lady A	1.0	0.5	0.0	
183	Ex For A Reason	Summer Walker & JT	Ex For A Reason	Summer Walker, JT & City Girls	1.0	1.0	0.0	
212	Rah Tah Tah	"Tyler, The Creator"	Rah Tah Tah	Tyler, The Creator	1.0	1.0	0.0	L
213	Work It	Missy "Misdemeanor" Elliott	Work It	Missy Elliott	1.0	1.0	0.0	
217	How Does It Feel To Be Forgotten	Selena Gomez & Benny Blanco	How Does It Feel To Be Forgotten	Selena Gomez & benny blanco	1.0	1.0	0.0	
232	Promotion	¥ : Kanye West & Ty Dolla ign Featuring Future	PROMOTION	¥ , Kanye West & Ty Dolla ign	1.0	1.0	0.0	
246	On BS	Drake & 21 Savage	On BS	Drake & 21 Savage	1.0	1.0	0.0	

Test 150 Songs

In [104...

```
test_songs = song_life[["Song", "Artist"]].sample(
    150,
    random_state=42
).copy()
```

In [106...

```
preview_results = []

for _, row in test_songs.iterrows():
    result = search_itunes_best_match(row["Song"], row["Artist"])
    preview_results.append(result)
    time.sleep(1.0)
```

```
preview_df = pd.DataFrame(preview_results)

preview_df["failure_reason"].value_counts()
```

```
Out[106]: failure_reason
SUCCESS    150
Name: count, dtype: int64
```

Testing 500 songs

```
In [82]: # test the Apple matching pipeline on 500 random Billboard songs
```

```
test_songs = song_life[["Song", "Artist"]].sample(
    500,
    random_state=42
).copy()

test_songs.head()
```

```
Out[82]:
```

	Song	Artist
33	"Girl, Get Up."	Doechii & SZA
1432	Bright	Echosmith
6779	Otis Jay Z Kanye West Featuring Otis Redding	
6901	Paramedic!	SOB X RBE
5211	Let It Go Keyshia Cole Featuring Missy Elliott & Lil Kim	

```
In [83]: # run
```

```
preview_results = []
failed = []

for _, row in test_songs.iterrows():

    result = search_itunes_best_match(
        row["Song"],
        row["Artist"]
    )

    # if Apple search fails completely, save the song for later inspection
    if result is None:
        failed.append((row["Song"], row["Artist"]))

    # if Apple returns a possible match, save the scored result
    else:
        preview_results.append(result)

    # slow down requests so we don't hammer the API
    time.sleep(0.2)

preview_df = pd.DataFrame(preview_results)
```

```
In [87]: # how many matched vs failed completely?
```

```
print("Matched:", len(preview_df))
print("Failed:", len(failed))
```

```
Matched: 60
Failed: 440
```

checks for after test:

```
In [80]: # match quality breakdown
```

```
preview_df["match_status"].value_counts()
```

```
Out[80]: match_status
AUTO_ACCEPT    49
REVIEW         7
LOW_CONFIDENCE  4
Name: count, dtype: int64
```

```
In [81]: # inspect random auto-accepted matches

preview_df[
    preview_df["match_status"] == "AUTO_ACCEPT"
][
    [
        "search_song",
        "search_artist",
        "matched_song",
        "matched_artist",
        "collection",
        "match_score",
        "title_score",
        "artist_score",
        "version_penalty",
        "match_status"
    ]
].sample(30, random_state=1)
```


Out [81]:

	search_song	search_artist	matched_song	matched_artist	collection	match_score	title_score	artist_score	ver:
32	Beg For It	Iggy Azalea Featuring M.O.	Beg For It (feat. M.O.)	Iggy Azalea	Reclassified	1.000000	1.0	1.000000	
39	New Romantics	Taylor Swift	New Romantics	Taylor Swift	1989 (Deluxe Edition)	1.000000	1.0	1.000000	
47	Fly As Me	Silk Sonic (Bruno Mars & Anderson .Paak)	Fly As Me	Silk Sonic, Bruno Mars & Anderson .Paak	An Evening With Silk Sonic	0.925000	1.0	0.750000	
57	Been You	Justin Bieber	Been You (Bonus Track)	Justin Bieber	Purpose (Deluxe)	1.000000	1.0	1.000000	
3	Paramedic!	SOB X RBE	Paramedic!	SOB X RBE	Black Panther: The Album	1.000000	1.0	1.000000	
4	Let It Go	Keyshia Cole Featuring Missy Elliott & Lil' Kim	Let It Go (feat. Missy Elliott & Lil' Kim)	Keyshia Cole	Just Like You	1.000000	1.0	1.000000	
50	Jail Pt 2	Kanye West	Jail pt 2	Kanye West	Donda	1.000000	1.0	1.000000	
34	Body Language	Jesse McCartney Featuring T-Pain	Body Language (feat. T-Pain)	Jesse McCartney	Body Language (feat. T-Pain) - Single	1.000000	1.0	1.000000	
53	"God, Your Mama, And Me"	Florida Georgia Line Featuring Backstreet Boys	God, Your Mama, And Me (feat. Backstreet Boys)	Florida Georgia Line	Dig Your Roots	1.000000	1.0	1.000000	
35	Made My Hoe Faint	Future	MADE MY HOE FAINT	Future	MIXTAPE PLUTO	1.000000	1.0	1.000000	
36	Bout A Million	Pop Smoke Featuring 21 Savage & 42 Dugg	Bout A Million (feat. 21 Savage & 42 Dugg)	Pop Smoke	Faith (Deluxe)	1.000000	1.0	1.000000	
46	Single Saturday Night	Cole Swindell	Single Saturday Night	Cole Swindell	Stereotype	1.000000	1.0	1.000000	
25	I Dreamed A Dream	Susan Boyle	I Dreamed a Dream	Susan Boyle	I Dreamed a Dream	1.000000	1.0	1.000000	
42	The Boys of Fall	Kenny Chesney	The Boys of Fall	Kenny Chesney	Hemingway's Whiskey (Deluxe Edition)	1.000000	1.0	1.000000	
23	This Ain't Nothin'	Craig Morgan	This Ain't Nothin'	Craig Morgan	That's Why	1.000000	1.0	1.000000	
49	Social Distancing	Lil Baby	Social Distancing	Lil Baby	My Turn (Deluxe)	1.000000	1.0	1.000000	
44	Motive	Ariana Grande Featuring Doja Cat	motive	Ariana Grande & Doja Cat	Positions	0.922857	1.0	0.742857	
31	I'm Still A Guy	Brad Paisley	I'm Still a Guy	Brad Paisley	Song Still Sings	1.000000	1.0	1.000000	
27	Just Say Dat	Gunna	just say dat	Gunna	The Last Wun	1.000000	1.0	1.000000	
15	Favorite Time Of Year	Carrie Underwood	Favorite Time of Year	Carrie Underwood	My Gift (Special Edition)	1.000000	1.0	1.000000	
48	Chill Bill	Rob \$toneFeaturingJ. Davi\$ & Spooks	Chill Bill (feat. J. Davi\$ & Spooks)	Rob \$tone	Chill Bill (feat. J. Davi\$ & Spooks) - Single	1.000000	1.0	1.000000	
20	Know No Better	"Major Lazer Featuring Travis Scott, Camila Ca...	Know No Better (feat. Travis Scott, Camila Cab...	Major Lazer	Know No Better - EP	1.000000	1.0	1.000000	
52	Electricity	Silk City x Dua Lipa	Electricity (feat. Diplo & Mark Ronson)	Silk City, Dua Lipa	Electricity (feat. Diplo & Mark	0.984211	1.0	0.947368	

	search_song	search_artist	matched_song	matched_artist	collection	match_score	title_score	artist_score	version_penalty
					Ronson) - Single				
29	Pumped Up Kicks	Foster the People	Pumped Up Kicks	Foster the People	Torches	1.000000	1.0	1.000000	
28	The Business	Yung Berg Featuring Casha	The Business (feat. Casha)	Yung Berg	Look What You Made Me	1.000000	1.0	1.000000	
5	I Wrote The Book	Morgan Wallen	I Wrote The Book	Morgan Wallen	One Thing At A Time	1.000000	1.0	1.000000	
37	The A Team	Ed Sheeran	The a Team	Ed Sheeran	+	1.000000	1.0	1.000000	
17	Having Our Way	Migos Featuring Drake	Having Our Way (feat. Drake)	Migos	Culture III	1.000000	1.0	1.000000	
12	The Sweet Escape	Gwen Stefani Featuring Akon	The Sweet Escape (feat. Akon)	Gwen Stefani	The Sweet Escape	1.000000	1.0	1.000000	
33	Who	Jimin	Who	Jimin	MUSE	1.000000	1.0	1.000000	

In [76]: *# inspect review matches, lowest confidence first*

```

preview_df[
    preview_df["match_status"] == "REVIEW"
][
    [
        "search_song",
        "search_artist",
        "matched_song",
        "matched_artist",
        "collection",
        "match_score",
        "title_score",
        "artist_score",
        "version_penalty",
        "match_status"
    ]
].sort_values("match_score").head(30)

```

Out [76]:

	search_song	search_artist	matched_song	matched_artist	collection	match_score	title_score	artist_
218	My Boo	Usher And Alicia Keys	My Boo	Alicia Keys & USHER	Confessions (Expanded Edition)	0.836364	1.000000	0.45
207	Tyrant	Beyonce & Dolly Parton	TYRANT	Beyoncé & Dolly Parton	COWBOY CARTER	0.838462	1.000000	0.46
26	Save Me	Jelly Roll With Lainey Wilson	Save Me (with Lainey Wilson)	Jelly Roll	Whitsitt Chapel	0.853846	1.000000	0.5
192	Our Kind Of Love	Lady Antebellum	Our Kind of Love	Lady A	Need You Now	0.871429	1.000000	0.51
188	Pray For Me	The Weeknd & Kendrick Lamar	Pray For Me	The Weeknd, Kendrick Lamar	Black Panther: The Album	0.871429	1.000000	0.51
110	Santa Baby	Eartha Kitt With Henri Rene And His Orchestra	Santa Baby (Remastered 2001)	Eartha Kitt	Purrfect - The Ultimate Collection	0.873684	1.000000	0.57
40	No Opp Left Behind	21 Savage & Metro Boomin	No Opp Left Behind	21 Savage & Metro Boomin	SAVAGE MODE II	0.874194	1.000000	0.58
137	No One Mourns The Wicked	"Ariana Grande Featuring Andy Nyman, Courtney-...	No One Mourns the Wicked (feat. Andy Nyman, Co...	Wicked Movie Cast & Ariana Grande	Wicked: The Soundtrack	0.877273	1.000000	0.59
106	Wild Child	Kenny Chesney With Grace Potter	Wild Child (with Grace Potter)	Kenny Chesney	The Big Revival	0.877273	1.000000	0.59
63	Fried	¥ : Kanye West & Ty Dolla ign	FRIED	¥ , Kanye West & Ty Dolla ign	VULTURES 2	0.881818	1.000000	0.60
237	You & Me	Gunna & Chloe	you & me	Gunna & Chlöe	DS4EVER	0.887500	1.000000	0.62
61	Girlfriend	Bow Wow & Omarion	Girlfriend	Bow Wow & Omarion	Girlfriend - Single	0.890909	1.000000	0.63
193	Smile	Juice WRLD & The Weeknd	Smile	Juice WRLD & The Weeknd	Smile - Single	0.893548	1.000000	0.64
85	Thrift Shop	Macklemore & Ryan Lewis Featuring Wanz	Thrift Shop (feat. Wanz)	Macklemore & Ryan Lewis	The Heist	0.893548	1.000000	0.64
2	Otis	Jay Z Kanye West Featuring Otis Redding	Otis (feat. Otis Redding)	Kanye West & JAY-Z	Watch the Throne	0.893548	1.000000	0.64
224	Boyfriend	Ariana Grande & Social House	Boyfriend	Ariana Grande & Social House	Boyfriend - Single	0.900000	1.000000	0.66
182	Grace	Lil Baby & 42 Dugg	Grace	Lil Baby & 42 Dugg	My Turn (Deluxe)	0.900000	1.000000	0.66
41	Hummingbird	Metro Boomin & James Blake	Hummingbird	Metro Boomin & James Blake	METRO BOOMIN PRESENTS SPIDER-MAN: ACROSS THE S...	0.900000	1.000000	0.66
66	I Luv Her	GloRilla & T-Pain	I LUV HER	GloRilla & T-Pain	GLORIOUS	0.908696	1.000000	0.69
43	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale	Kendrick Lamar & Tanna Leone	Mr. Morale & The Big Steppers	0.910000	1.000000	0.70
73	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain	Selena Gomez & The Scene	A Year Without Rain (Deluxe Edition)	0.911765	1.000000	0.70
125	Hate Me	Ellie Goulding & Juice WRLD	Hate Me	Ellie Goulding & Juice WRLD	Brightest Blue	0.915385	1.000000	0.71
198	Tomorrow 2	GloRilla & Cardi B	Tomorrow	GloRilla	Tomorrow - Single	0.927778	0.888889	1.00

In []:

additional

In []:

```
In [88]: # inspect failed searches
failed_df = pd.DataFrame(failed, columns=["Song", "Artist"])
failed_df.head(30)
```

Out[88]:

	Song	Artist
0	Anyways	Zach Bryan
1	Happy Xmas (War Is Over)	John Legend
2	Finally Found You	Enrique Iglesias Featuring Sammy Adams
3	Finesse	Drake
4	Ten Feet Tall	Afrojack Featuring Wrabel
5	Shoulda Let You Go	Keyshia Cole Introducing Amina
6	Girlfriend	Bow Wow & Omarion
7	Wishing	"DJ Drama Featuring Chris Brown, Skeme & Lyquin"
8	Working	Tate McRae X Khalid
9	Losers	Post Malone Featuring Jelly Roll
10	The Old Rugged Cross	Craig Wayne Boyd
11	Double Fantasy	The Weeknd Featuring Future
12	Me Too	Meghan Trainor
13	A Year Without Rain	Selena Gomez & The Scene
14	All My Life	Lil Durk Featuring J. Cole
15	I Know	YoungBoy Never Broke Again
16	Joe	Luke Combs
17	Young And Beautiful	Lana Del Rey
18	Yellow Flicker Beat	Lorde
19	In My Room	Frank Ocean
20	Let Me Go	Avril Lavigne Featuring Chad Kroeger
21	Psycho	Puddle of Mudd
22	When She Says Baby	Jason Aldean
23	You Got Em	Lil Durk
24	Keep Your Head Up	Andy Grammer
25	Thrift Shop	Macklemore & Ryan Lewis Featuring Wanz
26	For A Nut	Future Featuring Gunna & Young Thug
27	Rich Spirit	Kendrick Lamar
28	Ain't My Fault	Zara Larsson
29	Hold It Against Me	Britney Spears

```
In [89]: # test Apple directly on a few failed examples
for song, artist in failed[:10]:
    search_term_raw = f"{song} {artist}"
    search_term_clean = f"{clean_text(song)} {clean_text(get_main_artist(artist))}"

    print("RAW:", search_term_raw)
    print("CLEAN:", search_term_clean)

    response = requests.get(
        "https://itunes.apple.com/search",
        params={
            "term": search_term_clean,
            "media": "music",
            "entity": "song",
```

```

        "limit": 5
    }
)

data = response.json()
print("RESULT COUNT:", data.get("resultCount"))

if data.get("resultCount", 0) > 0:
    for r in data["results"][:3]:
        print("  -", r.get("trackName"), "|", r.get("artistName"))

print("-" * 80)

```

RAW: Anyways Zach Bryan
 CLEAN: anyways zach bryan

```

-----
JSONDecodeError                                Traceback (most recent call last)
File /opt/miniconda3/lib/python3.13/site-packages/requests/models.py:976, in Response.json(self, **kwargs)
    975 try:
--> 976     return complexjson.loads(self.text, **kwargs)
    977 except JSONDecodeError as e:
    978     # Catch JSON-related errors and raise as requests.JSONDecodeError
    979     # This aliases json.JSONDecodeError and simplejson.JSONDecodeError

File /opt/miniconda3/lib/python3.13/json/__init__.py:346, in loads(s, cls, object_hook, parse_float, parse_int, pars
e_constant, object_pairs_hook, **kw)
    343 if (cls is None and object_hook is None and
    344     parse_int is None and parse_float is None and
    345     parse_constant is None and object_pairs_hook is None and not kw):
--> 346     return _default_decoder.decode(s)
    347 if cls is None:

File /opt/miniconda3/lib/python3.13/json/decoder.py:345, in JSONDecoder.decode(self, s, _w)
    341 """Return the Python representation of ``s`` (a ``str`` instance
    342 containing a JSON document).
    343
    344 """
--> 345 obj, end = self.raw_decode(s, idx=_w(s, 0).end())
    346 end = _w(s, end).end()

File /opt/miniconda3/lib/python3.13/json/decoder.py:363, in JSONDecoder.raw_decode(self, s, idx)
    362 except StopIteration as err:
--> 363     raise JSONDecodeError("Expecting value", s, err.value) from None
    364 return obj, end

```

JSONDecodeError: Expecting value: line 1 column 1 (char 0)

During handling of the above exception, another exception occurred:

```

JSONDecodeError                                Traceback (most recent call last)
Cell In[89], line 19
     7 print("CLEAN:", search_term_clean)
     9 response = requests.get(
    10     "https://itunes.apple.com/search",
    11     params={
    (...)    16     }
    17 )
--> 19 data = response.json()
    20 print("RESULT COUNT:", data.get("resultCount"))
    22 if data.get("resultCount", 0) > 0:

File /opt/miniconda3/lib/python3.13/site-packages/requests/models.py:980, in Response.json(self, **kwargs)
    976 return complexjson.loads(self.text, **kwargs)
    977 except JSONDecodeError as e:
    978     # Catch JSON-related errors and raise as requests.JSONDecodeError
    979     # This aliases json.JSONDecodeError and simplejson.JSONDecodeError
--> 980     raise RequestsJSONDecodeError(e.msg, e.doc, e.pos)

```

JSONDecodeError: Expecting value: line 1 column 1 (char 0)

```

In [90]: def test_itunes_request(song, artist):
        """
        Debug one Apple/iTunes API request.
        Shows whether the issue is:
        - bad search term
        - no results
        - API/rate limit problem
        - JSON parsing problem
        """

```

```

search_term_clean = f"{{clean_text(song)}} {{clean_text(get_main_artist(artist))}}"

response = requests.get(
    "https://itunes.apple.com/search",
    params={
        "term": search_term_clean,
        "media": "music",
        "entity": "song",
        "limit": 5
    },
    timeout=10
)

print("SONG:", song)
print("ARTIST:", artist)
print("SEARCH TERM:", search_term_clean)
print("STATUS CODE:", response.status_code)
print("TEXT PREVIEW:", response.text[:200])

try:
    data = response.json()
    print("RESULT COUNT:", data.get("resultCount"))

    if data.get("resultCount", 0) > 0:
        for r in data["results"][:5]:
            print("-", r.get("trackName"), "|", r.get("artistName"))

except ValueError:
    print("JSON ERROR: Apple did not return valid JSON")

```

```

In [91]: for song, artist in failed[:5]:
        test_itunes_request(song, artist)
        print("-" * 80)
        time.sleep(1.0)

```

SONG: Anyways
ARTIST: Zach Bryan
SEARCH TERM: anyways zach bryan
STATUS CODE: 200
TEXT PREVIEW:

```
{
  "resultCount":5,
  "results": [
    {"wrapperType":"track", "kind":"song", "artistId":1436413980, "collectionId":1620257596, "trackId":1620257887, "artistName":"Zach Bryan", "collectionName":"America
RESULT COUNT: 5
→ If She Wants a Cowboy | Zach Bryan
→ Driving | Zach Bryan
→ This World's A Giant | Zach Bryan
→ 28 | Zach Bryan
→ Always Willin' | Zach Bryan
```

SONG: Happy Xmas (War Is Over)
ARTIST: John Legend
SEARCH TERM: happy xmas john legend
STATUS CODE: 200
TEXT PREVIEW:

```
{
  "resultCount":5,
  "results": [
    {"wrapperType":"track", "kind":"song", "artistId":16586443, "collectionId":1537876134, "trackId":1537876644, "artistName":"John Legend", "collectionName":"A Legend
RESULT COUNT: 5
→ Happy Xmas (War Is Over) | John Legend
→ Happy Xmas (War Is Over) [The Ultimate Mix] | John Lennon & Yoko Ono
→ Happy Xmas (War Is Over) [feat. The Harlem Community Choir] | John Lennon & The Plastic Ono Band
→ Happy Xmas (War Is Over) [feat. The Harlem Community Choir] [Alternate Mix] | John Lennon & Yoko Ono
→ Happy Xmas (War Is Over) [Remastered] | John Lennon, The Harlem Community Choir, The Plastic Ono Band & Yoko Ono
```

SONG: Finally Found You
ARTIST: Enrique Iglesias Featuring Sammy Adams
SEARCH TERM: finally found you enrique iglesias
STATUS CODE: 200
TEXT PREVIEW:

```
{
  "resultCount":5,
  "results": [
    {"wrapperType":"track", "kind":"song", "artistId":751069174, "collectionId":1892023918, "trackId":6761911228, "artistName":"Hartzon", "collectionName":"Finally Fou
RESULT COUNT: 5
→ Finally Found You | Hartzon
→ Finally Found You (Club Remix) | Flash Ki
→ Finally Found You | KIDZ BOP Kids
→ Finally Found You (128 BPM Workout ReMix) | Red Hardin
→ Finally Found You | Flash Ki
```

SONG: Finesse
ARTIST: Drake
SEARCH TERM: finesse drake
STATUS CODE: 200
TEXT PREVIEW:

```
{
  "resultCount":5,
  "results": [
    {"wrapperType":"track", "kind":"song", "artistId":271256, "collectionId":1418213110, "trackId":1418213392, "artistName":"Drake", "collectionName":"Scorpion", "trackName":"Finesse"
RESULT COUNT: 5
→ Finesse | Drake
→ Finesse | Drake
→ Finesse drake | Moneyybagg_blaze
→ Finesse | Pheelz & BNXN
→ Finesse | Bruno Mars
```

SONG: Ten Feet Tall
ARTIST: Afrojack Featuring Wrabel
SEARCH TERM: ten feet tall afrojack
STATUS CODE: 200

TEXT PREVIEW:

```
{
  "resultCount":5,
  "results": [
    {"wrapperType":"track", "kind":"song", "artistId":211210469, "collectionId":1443828382, "trackId":1443828610, "artistName":"AFROJACK", "collectionName":"Forget the
RESULT COUNT: 5
→ Ten Feet Tall (feat. Wrabel) | AFROJACK
→ Ten Feet Tall (feat. Wrabel) [David Guetta Remix] | AFROJACK
→ Ten Feet Tall (feat. Wrabel) / 2U (feat. Justin Bieber) [Afrojack Remix] [Mixed] | AFROJACK & David Guetta
→ Ten Feet Tall (feat. Wrabel) / Feels Like Home (Mixed) | AFROJACK & DubVision
→ Ten Feet Tall (feat. Wrabel) | AFROJACK
-----
```

Top40 Analysis

To focus on songs that achieved meaningful commercial success, audio analysis was restricted to songs reaching the Top 40 of the Billboard Hot 100.

This reduced the dataset from 10,505 songs to approximately 4,000 songs while retaining the most influential songs from each year.

```
In [143... top40["Song"].nunique()
```

```
Out[143... 3838
```

Pipeline Testing

```
In [144... top40 = song_life[song_life["peak_rank"] <= 40].copy()
```

```
In [145... test_top40 = top40[["Song", "Artist"]].sample(
    10,
    random_state=42
).copy()
```

```
In [146... preview_results = []

for _, row in test_top40.iterrows():
    result = search_itunes_best_match(row["Song"], row["Artist"])
    preview_results.append(result)
    time.sleep(1.0)

preview_top40_test = pd.DataFrame(preview_results)

preview_top40_test["artist_overlap"] = preview_top40_test.apply(
    lambda row: artist_overlap(row["search_artist"], row["matched_artist"]),
    axis=1
)

preview_top40_test["usable_match"] = (
    (preview_top40_test["match_status"] == "AUTO_ACCEPT")
    |
    (
        (preview_top40_test["title_score"] >= 0.95)
        & (preview_top40_test["artist_overlap"] >= 0.40)
        & (preview_top40_test["version_penalty"] == 0)
        & (preview_top40_test["preview_url"].notna())
    )
)

preview_top40_test[
    ["search_song", "search_artist", "matched_song", "matched_artist",
     "preview_url", "match_status", "usable_match"]
]
```


Out [146...	search_song	search_artist	matched_song	matched_artist	preview_url	match_status	usable_match
0	"No Body, No Crime"	Taylor Swift Featuring Haim	no body, no crime (feat. HAIM)	Taylor Swift	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
1	Stitches	Shawn Mendes	Stitches	Shawn Mendes	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
2	Homemade	Jake Owen	Homemade	Jake Owen	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
3	That's That	Snoop Dogg Featuring R. Kelly	That's That S*** (feat. R. Kelly)	Snoop Dogg	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
4	Beware	Big Sean Featuring Lil Wayne & Jhene Aiko	Beware (feat. Lil Wayne & Jhene Aiko)	Big Sean	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
5	Top Off	"DJ Khaled Featuring JAY Z, Future & B"	Top Off (feat. JAY-Z, Future & Beyoncé)	DJ Khaled	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
6	Decode	Paramore	Decode	Paramore	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
7	Just A Lil Bit	50 Cent	Just a Lil Bit	50 Cent	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
8	Chun-Li	Nicki Minaj	Chun-Li	Nicki Minaj	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True
9	Cupid	Fifty Fifty	Cupid	FIFTY FIFTY	https://audio-ssl.itunes.apple.com/itunes-asse...	AUTO_ACCEPT	True

```
In [148... # create folder for preview audio files
os.makedirs("preview_clips", exist_ok=True)
```

```
In [149... # grab first usable preview URL
test_row = preview_top40_test[
    preview_top40_test["usable_match"] == True
].iloc[0]

test_url = test_row["preview_url"]
test_filename = "preview_clips/test_preview.m4a"

test_url
```

```
Out[149... 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview221/v4/b8/61/cb/b861cb31-48eb-04d1-93c0-0d1209a734e
a/mzaf_206661796010024085.plus.aac.p.m4a'
```

```
In [150... # download one preview clip

response = requests.get(test_url)

print("Status code:", response.status_code)
print("Content type:", response.headers.get("Content-Type"))
print("File size:", len(response.content))

with open(test_filename, "wb") as f:
    f.write(response.content)

print("Saved to:", test_filename)

os.path.exists(test_filename)
```

```
Status code: 200
Content type: audio/x-m4p
File size: 1029713
Saved to: preview_clips/test_preview.m4a
```

```
Out[150... True
```

```
In [151... os.path.getsize(test_filename)
```

Out[151... 1029713

```
In [157... import essentia
import essentia.standard as es
```

```
In [153... import sys
print(sys.executable)
```

/opt/miniconda3/bin/python

```
In [154... !which python
```

/Users/trizzy/Personal/School/.venv/bin/python

```
In [155... !{sys.executable} -m pip install essentia
```

Collecting essentia

Downloading essentia-2.1b6.dev1389-cp313-cp313-macosx_15_0_arm64.whl.metadata (2.0 kB)

Requirement already satisfied: numpy>=1.25 in /opt/miniconda3/lib/python3.13/site-packages (from essentia) (2.3.5)

Requirement already satisfied: pyyaml in /opt/miniconda3/lib/python3.13/site-packages (from essentia) (6.0.2)

Requirement already satisfied: six in /opt/miniconda3/lib/python3.13/site-packages (from essentia) (1.17.0)

Downloading essentia-2.1b6.dev1389-cp313-cp313-macosx_15_0_arm64.whl (20.4 MB)

20.4/20.4 MB 31.3 MB/s eta 0:00:00m eta 0:00:010:0101

Installing collected packages: essentia

Successfully installed essentia-2.1b6.dev1389

```
In [158... loader = es.MonoLoader(filename=test_filename)
audio = loader()
```

```
print(type(audio))
print(audio.shape)
print(audio[:10])
```

<class 'numpy.ndarray'>

(1321920,)

[-0.14682572 -0.1478209 -0.14942724 -0.16521826 -0.16508675 -0.15349007
-0.16397864 -0.16048598 -0.12776649 -0.11213417]

```
In [159... duration_seconds = len(audio) / 44100
duration_seconds
```

Out[159... 29.975510204081633

```
In [160... # tempo (BPM)
```

```
rhythm = es.RhythmExtractor2013(method="multifeature")
bpm, beats, beat_confidence, _, _ = rhythm(audio)
```

```
print("BPM:", bpm)
print("Beat confidence:", beat_confidence)
```

BPM: 78.95964050292969

Beat confidence: 2.6579442024230957

```
In [161... # loudness
```

```
loudness = es.Loudness()

print("Loudness:", loudness(audio))
```

Loudness: 2235.899169921875

```
In [162... # key estimation
```

```
key_extractor = es.KeyExtractor()

key, scale, strength = key_extractor(audio)

print("Key:", key)
print("Scale:", scale)
print("Strength:", strength)
```

Key: Bb

Scale: major

Strength: 0.8196825981140137

```
In [163... # spectral centroid (brightness)
```

```
centroid = es.Centroid(range=len(audio))

print("Centroid:", centroid(audio))
```

Centroid: 668020.875

In []: